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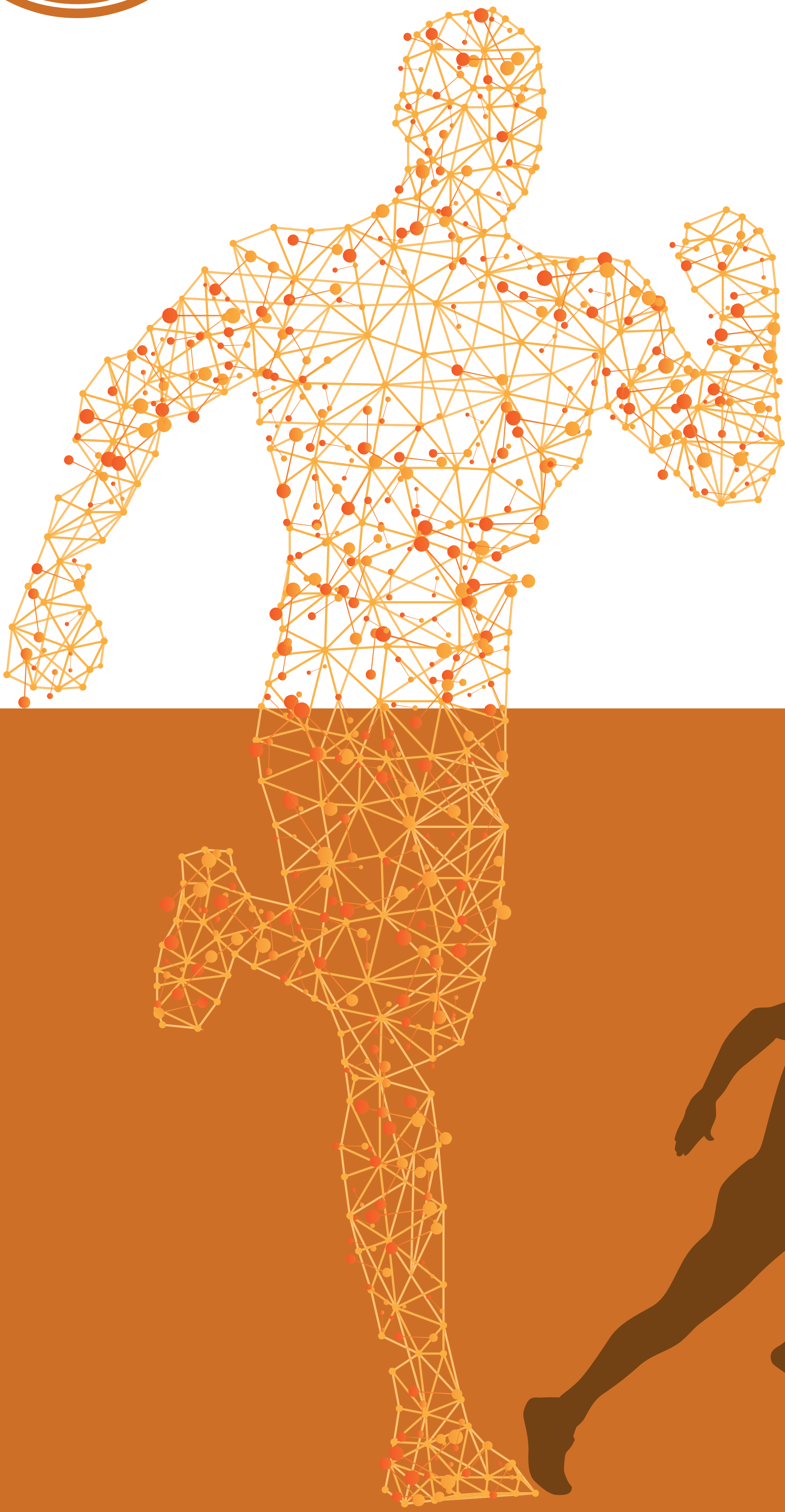

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Original Scientific Paper

Field Audit of Strength and Conditioning Coaches' Instructional and Motivational Language Repertoire

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Abstract

Strength and conditioning (S&C) coaches must employ psychological skills to optimally motivate athletes and promote their well-being. Yet, S&C coaches feel uncomfortable in their ability to apply such skills, highlighting a need for the development of science-based, practical tools. The purpose of this study was to examine the verbal language used by S&C coaches in publicly available YouTube videos through the Psychological Capital Model (PCM) lens. Coaches' statements (N = 178) were transcribed verbatim and coded into one of the eight dimensions of the PCM. Significant differences were found in S&C coaches' use of the eight developmental dimensions of the PCM, x^2 (7, N = 173) = 139.52, p < .0001, C = .67. Three PCM developmental dimensions were overused (i.e., standard residuals ranging from +2.76 to +7.10; i.e., experiencing success/modeling others [n = 54, 31.2%], building efficacy/confidence [n = 48, 27.8%], and implementing obstacle planning [n = 34, 19.7%]), while five were underused (i.e., standard residuals ranging from -2.23 to -4.18; i.e., building assets/avoiding risk [n = 11, 6.4%], persuasion and arousal [n = 10, 5.8%], affecting the influence process [n = 9, 5.2%]), goals and pathway design [n = 5, 2.9%]), and developing positive expectancy [n = 2, 1.2%]). To facilitate the use of a more diverse set of psychological strategies, this study offers a collection of 40 practice- and science-based motivational statements – five for each of the eight PCM dimensions – that S&C coaches may use and build upon to improve their own coaching language and practices.

Keywords: *Coaching Behavior, Exercise Leadership, Motivational Climate, Psychological Capital*

Introduction

Strength and conditioning (S&C) coaches play a unique and important role in the physical and personal development of student-athletes (Haff & Triplett, 2016). Due to the nature of the profession, S&C coaches spend a significant amount of time with their student-athletes during the academic year (Massey, Schwind, Andrews, & Maneval, 2009). Besides helping student-athletes improve their athletic performance in their respective sports and decrease the likelihood of injuries occurring through appropriate physical training and conditioning strategies, S&C coaches have the unique opportunity to foster student-athletes motivation and performance through the use of psychological skills (Radcliffe, Comfort, & Fawcett, 2013). Research among S&C coaches shows that motivation is one of the highest ranked critical factors influencing athletes' success, and a lack of motivation is the number one cause of poor performance (Radcliffe et al., 2013). Additionally, student-athletes perceive effective S&C coaches display effective communication, good listening, and motivational skills, as well

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as behaviors that promote trust, relatedness, and respect (Szedlak, Smith, Day, & Greenlees, 2015). Hence, what S&C coaches ultimately say or do can profoundly impact the relationship, effort, and performance potential of athletes (Cardinal & Melville, 1987).

Even though psychological knowledge and skills are viewed as foundational to success within S&C (National Strength and Conditioning Association, 2016; 2019), the extent to which S&C coaches develop such knowledge and skills and how those skills are used in practice is unclear. Within the S&C domain, most research has examined physical training strategies (Kraemer et al., 2017), with psychology-oriented research receiving comparatively little research attention (Radcliffe, Comfort, & Fawcett, 2013, 2018a, 2018b; Quartiroli, Moore, & Zakrajsek, 2020).

The aforementioned research to practice gap has been acknowledged by S&C professionals themselves. Specifically, they have expressed an interest in psychological skills and their development, yet they feel ill-prepared in their ability to employ such skills due to a lack of knowledge of different psychological strategies and a lack of confidence in using those strategies in practice (Radcliffe et al., 2018a). This is especially regrettable since Szedlak et al. (2015) identified S&C coaches' inspirational and motivational skills to be an essential component of their success in working with athletes.

As for the precise types of psychological skills S&C coaches are interested in, communication skills were ranked second highest in priority only behind hypnosis (Quartiroli et al., 2020). Coupled with the findings of Szedlak et al. (2015), whereby athletes strongly preferred their coaches use positive and encouraging language as a form of reward, there is a clear need for theoretically and empirically supported practical strategies that S&C coaches can implement into their practices (Moore & Gearity, 2019).

Though some attempts have occurred aimed at narrowing this gap, such as the "Psychology Special Issue" of the Strength & Conditioning Journal (Moore & Gearity, 2019), the results of these efforts have not always been immediately practical (e.g., introducing and overviewing theoretical models, concepts, and constructs; Schary, 2019). Rather, much of the available work tends to be conceptual or theoretical in nature versus based-in or derived-from practice (Statler & DuBois, 2016). That is, the S&C practitioner is often not only left to locate and decipher the work, but to also figure out how to put that work into practice. This can be a tall order even in more tangible areas of practice (Zenko & Ekkekakis, 2015). In part this is because S&C coaches work 64–75 hours per week (Massey et al., 2009). Thus, S&C coaches rely on other self-directed continuing education opportunities, such as online resources (Pope et al., 2015), which may help fill-in knowledge gaps.

Recognizing S&C coaches' desire, interest, and need for practical knowledge in the areas of communication, psychology, and motivation, and in an attempt to contribute to calls for more translational research that is grounded in theory and research within the discipline of kinesiology (Schary & Cardinal, 2016), the present study was undertaken. Specifically, S&C coaches' verbal language was reviewed within the context of the psychological capital model ([PCM], Harms & Luthans, 2012; Luthans, Avey, Avolio, Norman, & Combs, 2006). The PCM employs a human development perspective (e.g. motivation, psychology) by focusing on "who you are" and "who you are becoming" (Luthans et al., 2006). While the model does have empirical support, primarily through its application in industrial/organizational psychology, it has yet to be studied in the realm of S&C coaching. However, it has received some empirical attention within the realm of exercise leadership (Cardinal et al., 2015). In that study, exercise leaders in commercial exercise videos used language that was suspect in terms of building viewers' psychological capital.

Within the PCM there are eight developmental dimensions (i.e., goals and pathways design, implementing obstacle planning, experiencing success or modeling others, persuasion and arousal, building assets or avoiding risks, affecting the influence process, building efficacy or confidence, and developing positive expectancy). These developmental dimensions are hypothesized to be fostered or inhibited by the language used by S&C coaches (Luthans et al., 2006). Specifically, S&C coaches' language can be used to build: (1) hope through goals and pathway design and implementing obstacle planning, (2) efficacy through experiencing success/modeling others as well as persuasion and arousal, (3) resiliency through building assets/avoiding risks and affecting the influence process, and (4) optimism through building efficacy and confidence and developing positive expectancies.

In as much, the eight developmental dimensions are the conduits through which a S&C coach can inspire four hypothesized proximal outcomes. That is, hope, efficacy, resiliency, and optimism (HERO). Athletes who feel more hopeful, efficacious, resilient, and optimistic are more likely to take a positive approach to challenging tasks, as well as experience greater overall well-being (Luthans & Youssef-Morgan, 2017).

Unquestionably, a S&C coaches' communication style and motivational capabilities are related to athletes' psychological response, exercise behavior, and continuation of said behavior (Partridge, Knapp, & Massengale, 2014), thereby making the PCM a useful organizational framework for deconstructing and interpreting the motivational content being verbally conveyed by S&C coaches. Hence, the purpose of this study was twofold. First it was our aim to find publicly available S&C coaches YouTube videos at the college level and deconstruct and interpret the motivational content being verbally conveyed by S&C coaches. Second, we used the findings to create a catalog of phrases that encompass the full spectrum of developmental dimensions within the PCM.

Method

Experimental Approach to the Problem

This was an exploratory, cross-sectional, quantitative, content analysis study of the language derived from pre-recorded, public domain videos. The data obtained were extracted from these public artifacts and the researchers were not involved in the creation or posting of any of the videos. Due to the fact that these videos were public information shared in an online platform, research suggests that obtaining informed consent is not needed (Burles & Bally, 2018). As such, and because this work is unobtrusive, and no intervention or interaction with the individuals involved occurred, Institutional Review Board review was not required.

Sample

A purposive sample of ten "Mic'd up" videos were acquired vis-à-vis YouTube. These existing videos show insights into S&C coaches' daily work and interactions with athletes, as the coaches are wired with a microphone during various practice sessions. In an effort to obtain a range of interactions, we purposely sought variety in the sports teams being coached, as well as situations where male S&C coaches were working with male athletes (M-M) and male S&C coaches were working with female athletes (M-F). Videos displaying female S&C coaches working with athletes were almost nonexistent, and thus not included in the sample.

Individual members of the research team were assigned to locate different videos for possible inclusion in the study. A combination of words, such as "mic'd up", "college" "strength and conditioning", "male sports", and "female sports", were entered in the search area on YouTube to find potential videos. After the search, a total number of 20 videos were identified for possible use in this study. Videos that were exclusively/primarily interviews, philosophical, or promotional in nature were eliminated, as were videos depicting exclusively/primarily special occasion days or situations (e.g., maximal efforts). While such activities certainly fall within the scope of practice of S&C coaches, they likely encompass a very narrow range of instructional and motivational strategies. Essentially, we sought to identify videos that depicted the language and motivational strategies used in more routine, day-to-day activities, duties, and responsibilities that S&C coaches engage in. Against this backdrop, the research team reviewed all of the videos and collectively determined which ones to keep in and which ones to leave out based on the criteria mentioned above. For this analysis, five in the M-F category and five in the M-M category were retained, giving us a diverse and realistic set of observations from which to perform our analysis.

Measures and Procedures

The selected videos (N=10) were transcribed verbatim by the research team. In order to achieve consensus, a three-step process was employed during the video transcription. Each video was initially transcribed by one member of the research team. A second member then reviewed the initial transcript for accuracy and either verified or filled in potential gaps that were missed by the first transcriber. Lastly, a third member then independently reviewed and verified the transcript in order to confirm that all spoken words in the videos were accounted for and accurately recorded in the transcript. This three-step transcription, verification, and confirmation process resulted in full consensus among members of the research team.

The statements of each verified transcript were then reviewed regarding their motivational content by the entire five-person research team. Using the PCM, each statement was carefully interpreted and categorized into one of eight developmental dimensions, namely (1) goals and pathways design, (2) implementing obstacle planning, (3) experiencing success or modeling others, (4) persuasion and arousal, (5) building assets or avoiding risks, (6) affecting the influence process, (7) building efficacy or confidence, and (8) developing positive expectancy. The frequency of statements used was observed and calculated for each category. Each statement was only allowed to be classified into one of the eight dimensions.

Analysis

Descriptive statistics (e.g., M, SD, percentages, frequency counts) were computed for all variables. Given the nature and distribution of the data, as well as the overall sample size, primary analyses were carried out using non-parametric statistical tests (i.e., chi-square [χ^2], Mann-Whitney U [reported as z-scores]). Mean values were accompanied by Cohen's d computations, which were interpreted using the following guidelines: <0.20 = small, >0.20 to <0.80 = medium, and >0.80 = large (Cohen, 1988). As an adjunct to the χ^2-tests, contingency coefficients (C) were computed as a measure of magnitude. Contingency coefficient values >0.30 were considered substantial (Fleiss, 1981).

Results

The selected videos ranged in length from 1 minute to 6 minutes and 20 seconds (M = 2:04, SD = 1:35). They were published between 2012 and 2018 (M = 2015.70, SD = 1.94). The M-M vid-

TABLE 1. Psychological Capital Developmental Dimensions Employed by Strength and Conditioning Coaches.[a]

Variable	All Videos (N = 10)	Male-Male (M-M) Only Videos (n = 5)	Male-Female (M-F) Only Videos (n = 5)	Statistical Relationship (M-M vs. M-F)	Effect Size (Statistical Power)
Goals and Pathway Designs	M = 1.00 SD = 1.25 Range = 0-4	M = 1.40 SD = 1.52 Range = 0-4	M = 0.60 SD = 0.89 Range = 0-2	z-score = 0.836, p = 0.401	Cohen's d = 0.64 (.142)
Implementing Obstacle Planning	M = 0.50 SD = 0.97 Range = 0-3	M = 0.20 SD = 0.45 Range = 0-1	M = 0.80 SD = 1.30 Range = 0-3	z-score = -0.522, p = 0.603	Cohen's d = 0.62 (.136)
Experience Success/ Modeling Others	M = 5.40 SD = 5.10 Range = 0-15	M = 6.20 SD = 4.66 Range = 0-13	M = 4.60 SD = 5.94 Range = 1-15	z-score = 0.627, p = 0.529	Cohen's d = 0.30 (.061)
Persuasion and Arousal	M = 4.80 SD = 3.65 Range = 1-12	M = 4.80 SD = 4.44 Range = 1-12	M = 4.80 SD = 3.19 Range = 1-9	z-score = -0.209, p = 0.529	Cohen's d = 0.00 (N/A)
Building Assets/ Avoid Risk	M = 3.40 SD = 1.78 Range = 2-7	M = 2.80 SD = 1.30 Range = 2-5	M = 4.00 SD = 2.12 Range = 2-7	z-score = -0.731, p = 0.465	Cohen's d = 0.68 (.155)
Affecting the Influence Process	M = 0.20 SD = 0.42 Range = 0-1	M = 0.20 SD = 0.45 Range = 0-1	M = 0.20 SD = 0.45 Range = 0-1	z-score = 0.104, p = 0.920	Cohen's d = 0.00 (N/A)
Building Efficacy/ Confidence	M = 1.10 SD = 1.85 Range = 0-6	M = 0.40 SD = 0.55 Range = 0-1	M = 1.80 SD = 2.49 Range = 0-6	z-score = 0.836, p = 0.401	Cohen's d = 0.78 (.191)
Developing Positive Expectancy	M = 0.90 SD = 1.85 Range = 0-6	M = 0.40 SD = 0.55 Range = 0-1	M = 1.40 SD = 2.61 Range = 0-6	z-score = 0.104, p = 0.920	Cohen's d = 0.53 (.111)
Total Number of Observations	N = 178 (100%)	n = 82 (47.40%)	n = 91 (52.60%)	z-score = -0.465, p = .638	Odds ratio = 1.11 (N/A)

[a] Unless otherwise specified, values shown are average number of occurrences per video.

eos (n = 5) included one from basketball, three from American football, and one from ice hockey. The M-F videos (n = 5) included three from basketball, one from softball, and one from volleyball. A total of 178 statements were recorded. On average, M-M videos had twice as many views (M = 4,926.80, SD = 8,433.10) as M-F videos (M = 2,395.60, SD = 2,919.89), yielding a medium effect size (i.e., d = 0.40).

As illustrated in Table 1, no significant differences were observed regarding the use of psychological capital developmental dimensions between S&C coaches working with female or male athletes. Though not statistically significant, the observed effect sizes suggest some emergent differences in terms of the average usage of the developmental dimensions of the PCM. Coaches working with female athletes used more building efficacy and confidence statements (M = 1.80, SD = 2.49) than coaches working with male athletes (M = 0.40, SD = 0.55), U = 0.836, p = 0.401, d = 0.78. The same trend was observed for motivational statements focusing on overcoming obstacles, building assets/avoiding risk, and developing positive expectancy. In contrast, coaches working with male athletes used more goal setting statements (M = 1.40, SD = 1.52) compared to coaches working with female athletes (M = 0.60, SD = 0.89), U = 0.836, p = 0.401, d = 0.64.

When looking at S&C coaches' use of the eight developmental dimensions of the PCM, significant differences between the dimensions were found χ^2 (7, N = 173) = 139.52, p < .0001, C = .67 (see Table 2). Three out of the eight dimensions, including experiencing success/modeling others (n = 54, 31.2%), building efficacy/confidence (n = 48, 27.8%), and implementing obstacle planning (n = 34, 19.7%), accounted for 136 (76.4%) of the total statements. The analysis revealed that these statements were overrepresented (i.e., standard residuals ranging from +2.76 to +7.10). Respectively, these statements are hypothesized to develop efficacy, optimism, and hope. The other five dimensions, including building assets/avoiding risk (n = 11, 6.4%), persuasion and arousal (n = 10, 5.8%), affecting the influence process (n = 9, 5.2%), goal and pathways design (n = 5, 2.9%), and developing positive expectancy (n = 2, 1.2%), were underrepresented (i.e. standard residuals ranging from -2.23 to -4.18).

TABLE 2. Observed Versus Expected Use of Psychological Capital Developmental Dimensions by Strength and Conditioning Coaches (N = 10).

Rank	Developmental Dimension (Hypothesized Proximal Outcome)	Observed Frequency	Expected Frequencya	Expected Proportiona	Actual Proportion	Percentage Deviation	Standardized Residuals
1	Experiencing Success/Modeling Others (Efficacy)	54	21.625	.125	.312	+153.88%	+7.10
2	Building Efficacy/Confidence (Optimism)	48	21.625	.125	.278	+125.67%	+5.80
3	Implementing Obstacle Planning (Hope)	34	21.625	.125	.197	+59.85%	+2.76
4	Building Assets/ Avoiding Risk (Resiliency)	11	21.625	.125	.064	-48.28%	-2.23
5	Persuasion and Arousal (Efficacy)	10	21.625	.125	.058	-52.99%	-2.44
6	Affecting the Influence Process (Resiliency)	9	21.625	.125	.052	-57.69%	-2.66
7	Goals and Pathway Design (Hope)	5	21.625	.125	.029	-76.49%	-3.53
8	Developing Positive Expectancy (Optimism)	2	21.625	.125	.012	-90.60%	-4.18

[a] Hypothetical, assumes equal development of all psychological capital dimensions, χ^2 (7, N = 173) = 139.52, p < .0001, C = .67

Discussion

In this study, S&C coaches verbal motivational language was coded on the basis of the PCM. The interactions of 10 male S&C coaches working with athletes in men's basketball, American football, and ice hockey, and women's basketball, softball, and volleyball were observed. Interactions were classified on the basis of S&C coaches use of the eight developmental dimensions of the PCM. These eight developmental dimensions are the pathways through which an athlete's sense of hope, efficacy, resiliency, and optimism are fostered. On the basis of observed effect sizes, coaches working with female athletes seemed to use more building efficacy and confidence statements, as well as statements that focused on overcoming obstacles, building assets/avoiding risk, and developing positive expectancy than did coaches working with male athletes. Contrary, coaches working with male athletes seem to use more goal setting statements than did coaches working with female athletes. These observations should be interpreted with caution since statistical significance was not established. However, statistically significant differences were found regarding the overall usage of the eight developmental dimensions of the PCM. Three out of the eight dimensions, namely experiencing success/modeling others, building efficacy/confidence, and implementing obstacle planning, were overused, while the remaining five were relatively underused. Two of the five least used statements by S&C coaches are hypothesized to develop resiliency (i.e., building assets/avoiding risk and affecting the influence process), which is certainly a critical element of athletic and life success.

The developmental dimensions of the PCM are the conduits through which S&C coaches are hypothesized to build four proximal outcomes in athletes, namely hope, efficacy, resiliency, and optimism. Each proximal outcome will next be discussed in the context of the study's findings.

Hope

In the present study, S&C coaches only partially used language that fosters a sense of hope among their athletes. This is disappointing because hope is a cognitive process that gives one a sense of determination to accomplish a goal (Snyder, 2000). Even when controlling for other factors, such as self-esteem, athletes with high hope display greater academic and athletic performances (Curry & Snyder, 2000).

The majority of hope-building statements used by S&C coaches in this study focused on implementing obstacle planning, including statements such as, "I don't care what everyone else is saying, you just talk to yourself and go." By contrast, they infrequently used statements aimed at fostering hope that focused on goal and pathways design-type statements, such as, "We are going to do one rep for the following reason…" Statements such as this have been described as essential characteristics of expert S&C coaches (LaPlaca & Schempp, 2020).

Efficacy

Experiencing success/modeling others was the most frequently used dimension, which instills efficacy or a sense of confidence in one's own abilities. High self-efficacy is associated with showing greater effort, perseverance, and seeking out challenging tasks (Bandura, 1986). The importance of self-efficacy in sports performance is well established (Moritz, Feltz, Fahrbach, & Mack, 2000), including its value in developing positive muscular fitness promoting behaviors (Cardinal & Kosma, 2004). As such, coaches are encouraged to foster athlete's self-efficacy to help them increase their sports training and performance. A source of building self-efficacy is verbal persuasion (Bandura, 1986), which can be accomplished through positive encouragement and verbal reinforcement. S&C coaches in our sample frequently used statements, such as "good job", "there you go" or "solid work". Contrary, persuasion and arousal statements, such as "Let's get it done, come on now", "Ready?" and "Explode up!" were used infrequently, even though these statements would have similar effects on building athlete's efficacy. This indicates that coaches often use praise-type language surrounding verbal reinforcement, which are easy to use in order to nurture athletes' efficacy. Yet, overused praise and verbal reinforcements can become meaningless to athletes very quickly, if these comments are vague, deconstructive, and/or insincere (Huber, 2013). Acquiring knowledge and strategies on how to use specific and efficient language in order to increase athlete's efficacy is an important skill to develop and master for S&C coaches.

Resiliency

The ability to withstand pressure and bounce back from adversity is known as resiliency (Sarkar & Fletcher, 2014). Challenging situations are commonplace during training, competition, and in life, therefore it is crucial for athletes to develop resilient behaviors and skills. S&C coaches are uniquely positioned to help develop such behaviors and skills among their athletes, especially as it pertains to preventing and overcoming sports injuries (Talpey & Siesmaa, 2017). However, in the present study, S&C coaches underused language aimed at fostering resiliency relative to the other outcomes (i.e., efficacy, optimism, and hope). Example statements for building resiliency include, "Attack the rep now, attack the rep, let's go" or "I know you had a rough night last night; I know it was tough yesterday. I know it was tough the day before that. But that's how it is. It's a grind." Such statements were scarcely used, which creates an opportunity for improvement in terms of coaching behaviors.

Optimism

Building efficacy and confidence can cultivate one's optimism, which is a person's expectation for positive future outcomes. Optimism is associated with greater sports performance (Ortin-Montero, Martínez-Rodríguez, Reche-García, de los Fayos, & González-Hernández, 2018), which underscores the importance of S&C coaches exercising strategies that increase optimism among the athletes in their charge. In our sample, coaches frequently used language such as "You are stronger than you think" or "Who knew you could squat that low, huh?" Such statements can foster optimism because they help build efficacy and confidence. In contrast, building optimism through the development of positive expectancy was the least used of the eight developmental dimensions. Nevertheless, statements, such as "If you can learn to work through it now, it will be easier in the game. I promise you" are helpful in instilling positive expectations in one's abilities. S&C coaches are encouraged to work on developing this strategy, as it will help build an optimistic outlook among their athletes.

Limitations

This study has several limitations. These include the following. First, the 10 "Mic'd up" videos were acquired purposively. Second, only the verbal motivational languages of S&C being featured in the videos were assessed. No attempt was made to assess their nonverbal behavior or the verbal or nonverbal of others who were present (e.g., spotters, training partners). It is unclear how much the videos were edited, rehearsed, or scripted prior to their distribution. Regardless, the videos were deemed to be authentic and representative by the two members of the research team who are active as S&C coaches, one former S&C coach, and all members of the research team who were former collegiate level athletes. Finally, only male S&C coaches were assessed in this study. Future work may assess female strength and conditioning coaches' verbal motivational language.

Practical Applications

Tangible Tool for Building Athletes' Psychological Capital

While the importance of language and cueing in coaching from a motor learning and skill acquisition perspective has been written about (Winkelman, 2021), other practice-oriented resources that focus on psychological and motivational tools continue to be scarce. This is part of a larger problem, which is the challenge professionals face in putting research and theory into practice (Knudson, 2005; Schary & Cardinal, 2015). This problem has been observed among S&C coaches in particular (Eisenmann, 2017).

As observed in the present study, several of the developmental dimensions of the PCM were underused by S&C coaches. In an attempt to narrow the research-into-practice gap, as well as address the full range of motivational strategies in the PCM, a tangible outcome of this study was the development of an inventory of 40 practical coaching statements that are theoretically in alignment with each of the eight developmental dimensions of the PCM (i.e., five statements per dimension; see Table 3). The collection of statements represents acquired quotes from the videos and quotes that stemmed from the videos but were modified to fulfill the developmental dimension. S&C coaches are encouraged to rehearse and try out the full repertoire of statements in their everyday coaching practices. These examples can be used to elicit new ideas on how to phrase motivational language during practice and/or competition and should be changed or adjusted according to the specific context and student-athlete. Specific strategies may intentionally be employed on specific days or occasions. Early career professionals, including those doing internships or practicums, may use this inventory to develop unique skills in the area of communication and psychology (Martin, 2020).

TABLE 3. Psychological Capital Development Model: Conceptual Overview of the Developmental Dimensions, Specific Examples, and Hypothesized Outcomes

Developmental Dimensions (i.e., how the strength and conditioning coach seeks to develop her/his athletes)	Examples of Each Developmental Dimension as Used in Practice. (Note: Quoted material acquired directly from the videos reviewed.)	Proximal Outcomes (i.e., hypothesized psychological capital being developed in the athletes as a result of the coach's actions)
Goals and Pathway Designs (e.g., challenge and/or goal setting)	1. "We all must come across the line at 65 because that's what it's going to take to do what we are trying to do which is win the championship, plain and simple." 2. "We are going to do one rep for the following reason: …" 3. "Last one – down and up!" 4. "This is about the time you want to slow down. Don't slow down!" 5. On the last set I want you at 275.	Hope (i.e., fostering a belief that participants can accomplish their goals)
Implementing Obstacle Planning (e.g., getting over obstacles, such as offering modifications; fostering a belief that participants are in control of their lives)	1. "You can't do it by yourself [insert name]. If you could, s/he wouldn't be over there helping you." 2. "Give everything you have to finish. Finishing, that's you." 3. "I don't care if you listen to what anyone else is saying, you just talk to yourself and you go." 4. "When you decide to, things happen. So just decide to." 5. Keep your form. Let your spotters' help you finish with good form.	Hope (i.e., fostering a belief that participants can accomplish their goals)
Experience Success/Modeling Others (e.g., reinforcement and/or celebratory comments)	1. "Really good job in the weight room – really good job out here." 2. "There you go!" 3. "Solid work!" 4. "That's how you finish!" 5. That's what we like to see. That's what we are looking for.	Efficacy (i.e., fostering confidence in participants' abilities or assuring a sense of confidence in their abilities)
Persuasion and Arousal (e.g., check-ins, stimulation)	1. "Let's get it done, come on now." 2. "Let's go, hustle up. On the move. Get there." 3. "Everybody got me?" "Any questions, anything?" 4. "Explode up!" 5. Did that feel better? "You feel the difference?"	Efficacy (i.e., fostering confidence in participants' abilities or assuring a sense of confidence in their abilities)
Building Assets/Avoid Risk (e.g., enthusiasm, establishing an exciting, fun, environment)	1. "Go somewhere. Buzz around it. Downhill! Sprint! Down and ready!" 2. "Now go get him. Downhill." 3. "Attack the rep now, attack the rep, let's go!" 4. "You've got to lock into it!" 5. Let's go! Be aggressive. Show me what it takes!	Resiliency (i.e., fostering a belief that participants can bounce-back from setbacks that may occur.
Affecting the Influence Process (e.g., growth mindset; persist despite obstacles; recognize effort as the pathway to success; learn from criticism)	1. "I know you had a rough night last night. I know it was tough yesterday. I know it was tough the day before that. But that's how it is. It's a grind." 2. "Always dropping our head when stuff get tough. Stand tall. Look your opponent in the eye." 3. Heads up. Tomorrow's another day. 4. We struggled in that speed session. We need to have a good lift in the weight room. 5. The effort is there. It will come. We are on the right path.	Resiliency (i.e., fostering a belief that participants can bounce-back from setbacks that may occur.
Building Efficacy/Confidence (e.g., encouragement of any type; i.e., both effort and results)	1. "You're stronger than you think!" 2. "Push it. Push it. See it through [insert name]. See it through [insert name]." 3. "Who knew you could squat that low, huh?" 4. You got one more. You can do this. Common [insert name]. 5. I wouldn't let you do anything I didn't think you could do.	Optimism (i.e., fostering an expectation that good things will happen to the participants in the future

(Continued from previous page)

Developmental Dimensions (i.e., how the strength and conditioning coach seeks to develop her/his athletes)	Examples of Each Developmental Dimension as Used in Practice. (Note: Quoted material acquired directly from the videos reviewed.)	Proximal Outcomes (i.e., hypothesized psychological capital being developed in the athletes as a result of the coach's actions)
Developing Positive Expectancy (e.g., Expectancy, anticipation, anticipatory set)	1. "The championship is on the line. They got you two times in regular season and now it's on the line. Winner take all; winner take all. Right here right now. Win the rep, win the game, win the title. Plain and simple." 2. "Now is when I am teaching you how to learn to work through fatigue. If you can learn to work through it now, it will be easier in the game. I promise you. It's easy in a game." 3. So if s/he pushes you in the back or s/he pushes in the shoulder, wherever s/he pushed you to assist you, you take it until you decide to do it on your own. It's called assistance. It's called teamwork. It's called buddying-up. It's called accountability. All the things necessary to win championships. [Modified] 4. "This is 85%. You should smoke this." 5. I've got to see more purpose and passion in what you are doing.	Optimism (i.e., fostering an expectation that good things will happen to the participants in the future

Note: Statements with quotation marks are direct quotes from the videos reviewed, whereas statements without quotations marks were developed by the research team.

Conclusion

Sport and exercise psychology research specific to S&C settings continues to receive attention; yet, rarely does it offer practice-oriented implementation strategies. The current study highlights the restricted range of psychological strategies used by S&C coaches. Three of the eight developmental dimensions of the PCM were used frequently, partially targeting efficacy, optimism, and hope outcomes. However, five of the dimensions were underused, mainly illustrating the insufficient use of language targeting resiliency relative to the other outcomes. Given the limited range of motivational language observed in the videos, this study provides a tangible list of 40 motivational statements in an effort to expand S&C coaches' motivational strategies repertoire. By providing these practice- and science-based motivational statements, this study also attempts to narrow the gap between theory, research, and practice.

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Conflict of Interest
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Original Scientific Paper

The Attitudes of Montenegrin Billiard Players towards Necessity to Establish Billiard Association

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Abstract

In Montenegro billiards has it's long history, but according to fact that it is actively and en masse played throughout Montenegro, it certainly has it's present and future. Aim of this research is to examine attitudes of billiard players from Montenegro towards necessity to establish billiard association. The instrument of this research is a survey questionnaire. The respondents are active billiard players from Montenegro, 78 of them, average age of 33.35±7.94 years, who expressed their attitudes by choosing one of offered answers on asked question. This study results are analysed by Google Forms platform. Based on the results of this research, the conclusion is that Montenegrin billiard players consider that Montenegro needs to establish national billiard association.

Keywords: *Attitudes, Snooker, Game, Billiard Association, History of Billiards, Montenegro*

Introduction

In Montenegro, billiards appears for the first time in 1836 when Petar II Petrovic Njegos back then brought first and only billiard table from Vienna to Cetinje (Montenegro Travel, n.n.). There are several types of billiards in the world, most common are snooker and pool, and within pool there are four games – eight ball, nine ball, ten ball and straight pool (Elmaged, 2017). Nowadays in Montenegro are played first three aforementioned pool games and snooker is on the rise. Period of prosperity of this sport from organizational level happened in period of existence of billiard association of Montenegro, from 1996 to 2011, but after that there was a stagnation period which still lasts (Radoicic, Milosevic, Zarkovic, & Masanovic, 2021). In personal communication with founder and president of former association of Montenegro, Mr Dragan Scepanovic, we collected data about existence and functioning of former association. In 1996, Dragan Scepanovic founded the Billiards Association of Montenegro in accordance with the then law on the establishment of a sports organization. According to his words, first competitive year under the patronage of Montenegrin billiard association (league, individual championship and masters) was held in 1999. Under the patronage of former association, there were eight clubs from Podgorica, Bijelo Polje, Niksic, Spuz, Danilovgrad, Tivat. Mr Scepanovic says that to all billiard players from Montenegro billiard was an activity they practiced in leisure, so they didn't earn for living by playing billiards. The junior league was held in season 2001/2002 and then there was a billiard course within billiard club "Podgorica". The female league was held in the season 2004/2005. At competitions beyond Montenegrin borders, billiard players had much success at tournaments of former federal league (Federal Republic of Yugoslavia and later State Union of Serbia and Montenegro), conquering first, second and third places. Montenegrin national team competed at several European championships without significant results. Mr Scepanovic highlighted that billiard club "Podgorica" during whole period of existence of Montenegrin billiard association organized well known tournament "Montenegro Open" 11 years in row in period 1999-2010, which was the most famous tournament in the region, namely, in area of former Yugoslavia. Montenegrin billiard association stopped existing in 2011, when registration expired and billiard club "Podgorica", which was base for all competitions within association, was closed.

In personal communication with Mr Danijel Garic, initiator for establishment of Monetengrin billiard association and future president of one, we gathered information about current state of

Montenegro Sport

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billiards in Montenegro and future aspirations. He highlighted that in Montenegro, there are organized and well-equipped clubs in all major cities in all three regions (south, central, north), that even without patronage of national organization new, modern clubs are opening, where is spent a lot of time in order to improve game and training process, as well as to educate young players. Mr Garic points out that billiard centers have their municipal leagues, organized voluntarily, they are massive, regular and very well organized, that tournament calendar is densely filled and that participants are enthusiastic and paying their own costs for arriving and participating. Despite everything, clubs are able to organize even the most demanding regional tournaments with participants who are the best players from Balkans, who have the best impressions about tournament organization and always praise it. Collaboration between clubs in Montenegro is impeccable, tournaments beside competitive have also friendly character so even without aforementioned non-existence of patronage association, within that without referee organization, rules on tournaments are respected and matches are played in spirit of fair play and without any incidents. He further states that our country needs national billiard association which would patronage all vital aspects of billiard games (competition organization, club functioning, education of young people and representation of Montenegrin billiard). Plans for establishment of association paused for few years, are finally intensified during last year, at the beginning of 2020 those were discussed with former minister of sport and president of Montenegrin Olympic Committee, who welcomed initiative and promised their required help during registration of national association. Further activities, such as required registration of clubs needed for association establishment, were unexpectedly stopped by pandemic caused by Covid 19 virus, so they are still a current issue. At the end, Mr Garic added that Montenegrin billiard players are promising and that they deserve to have support of national association.

According to aforementioned, aim of this research is to publicly present attitudes of billiard players from Montenegro towards necessity to establish billiard association, population which based on enthusiasm achieves great results despite support absence of an umbrella organization. All of this with want to popularize this sport and to provide players, who are untiring in preserving tradition of this sport in Montenegro, to build stabile future.

Methods

The population of this retrospective cross-sectional study includes 78 active billiard players from Montenegro, average age of 33.35±7.94 years. All respondents are adults with permanent residence in Montenegro who actively play billiard (most of them for more than 5 years). During sample selection it was taken into account that distribution is divided into as many Montenegrin municipalities as possible. It is necessary to remind that all participants participated voluntarily in this study and they had ability to resign their participation in this research at any point.

The instrument of this research is a survey questionnaire which contains seven questions divided in two subsystems. First three questions from questionnaire are related to attitudes of Montenegrin billiard players towards former association (i was informed about the existence and activity of the former billiard association; the organization of internal competitions was better during the activities of the former billiards association; the participation in international competitions was more frequent during the activities of the former billiards association). Last four questions from questionnaire are related to attitudes of Montenegrin billiard players towards necessity to establish billiard association and changes that it would cause (it is necessary for Montenegro to establish a billiards association; establishing a billiards association would increase participation in major competitions; establishing a billiards association would encourage young people to play billiards; by establishing billiard association successful players will get the status of a professional player). This questionnaire was created by this research author.

Research questionnaire was conducted via Google Forms platform in period October 10th to 24th in 2020. Questions were closed with offered answers yes or no. It is important to point out that survey was anonymous and that all answers were highly confidential. Also, this study author precisely inspected and controlled, that is, rejected all survey questionnaire answers which were not neatly filled, and there were eight of them.

All data in this research, collected by filling out a questionnaire in the Google Forms survey by Montenegrin billiard players, were directly exported and summarized in the Google spreadsheet in percentages.

Results

Based on firtst subsystem answers (attitudes of Montenegrin billiard players towards existence and functioning of former billiard association) it is noticeable that: familiarity to existence and functioning of former billiard association on level of Montenegrin billiard players is in half (Figure 1); most of the players consider that competition organization in country was better back in time of existence of former billiard association (Figure 2); most of the players consider that participation at international competitions was more frequent in period of existence of former billiard association (Figure 3).

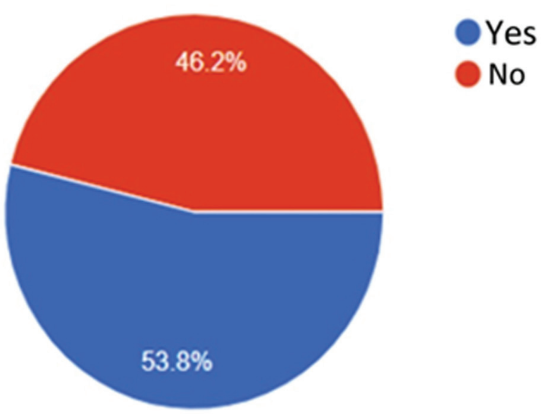

FIGURE 1. I was informed about the existence and activity of the former billiard association

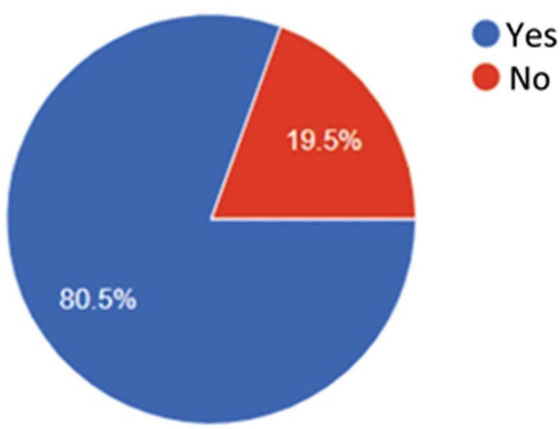

FIGURE 2. The organization of internal competitions was better during the activities of the former billiards association

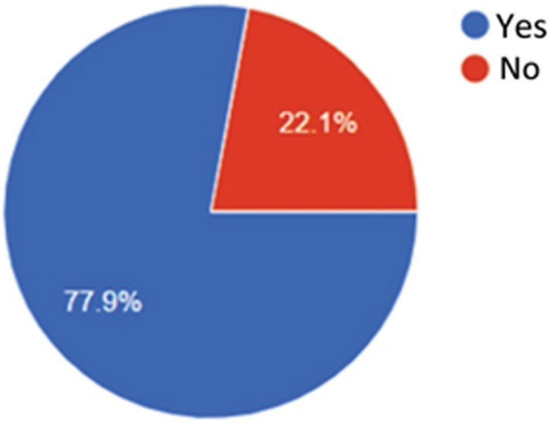

FIGURE 3. Participation in international competitions was more frequent during the activities of the former billiards association

Based on second subsystem answers (attitudes of Montenegrin billiard players towards establishment of billiard association and their perception of advatntages which that change could provide) it is noticeable that: Montenegrin billiard players almost unanimously consider that Montenegro needs billiard association (Figure 4); they are unanimous with attitude that billiard association establishment would provide them participation at bigger competitions (Figure 5); almost all players consider that billiard association establishment would attract more young people to take this sport seriously (Figure 6); absolute majority of Montenegrin billiard players is unique in attitude that by establishment of billiard association successful players would have professional sportsman status (Figure 7).

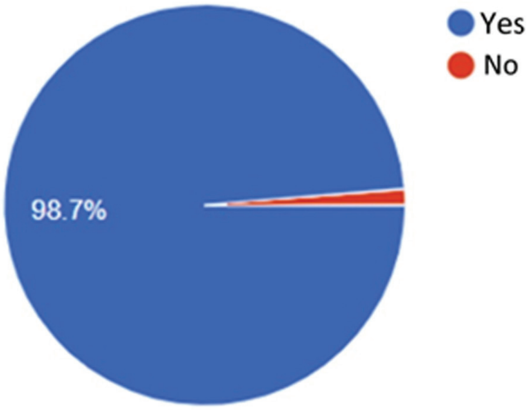

FIGURE 4. It is necessary for Montenegro to establish a billiards association

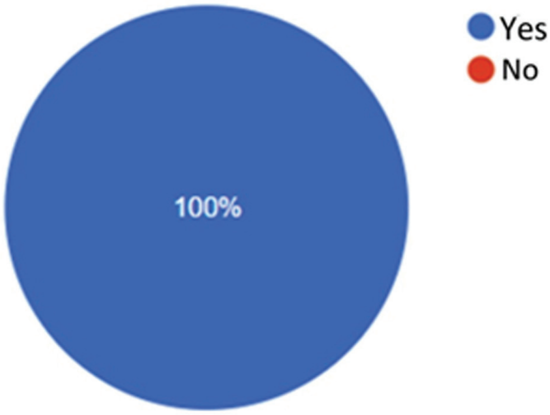

FIGURE 5. Establishing a billiards association would increase participation in major competitions

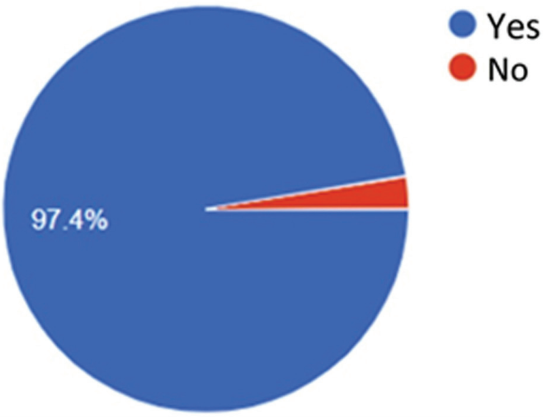

FIGURE 6. Establishing a billiards association would encourage young people to play billiards

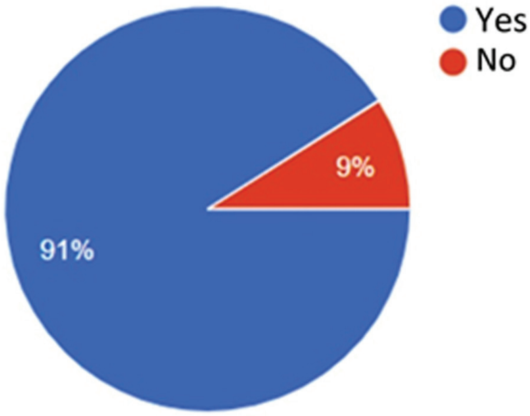

FIGURE 7. By establishing billiard association successful players will get the status of a professional player

Discussion

This study results clearly demonstrate that Montenegrin billiard players give the impression of very organized group gathered with aim to improve billiards status in Montenegro. Considering for how long without an umbrella organization they achieve to successfully organize and behave professionally, it is concluded that huge passion about billiards gathers them in it. Fact that billiard has extremely long tradition in Montenegro, where it is played since 1836 (Montenegro Travel, n.n.) made an impact on that for sure. However, after 2011 when Montenegrin billiard association activity stops, all the way until today, this sport evidently was in some kind of crisis, without support. The question is would future activities of billiard community be successful and provide necessary support which would enable establishment of billiard association.

Results from first questionnaire subsystem show that Montenegrin billiard players aren't informed enough about existence and functioning of former billiard association (46.2% of players doesn't have enough information). Reason for that, among others, is lack of written and published data about this topic for sure, and this research will surely help to replace lack of written materials and knowledge about one significant part of history of Montenegrin billiards. Beside ignorance in former billiard association functioning, most of the players are informed from older colleagues that competition organization was better back in that period (80.5%) and that international competition participation was more massive (77.9%), which means that they are aware that necessity of association establishment is undeniable. It is important to highlight that knowledge of facts about existence and functioning of former association may be helpful for future leaders of new billiard association if it would be established, that based on previous experience use some quality works in their future plan, to use it as an inspiration for developing plan (existence of junior and female league), and that future association sets healthy foundation which could provide its long existence. Beginning may be hard and challenging, but none of the steps should be skipped, because nobody wants to have reestablished billiard association which would last shortly. In order that this sport develops and begin to live in Montenegro, continuity of existence of billiard association or an umbrella organization is necessary as support to quality players.

Results of second questionnaire subsystem show that Montenegrin billiard players have positive and unanimous attitude towards necessity to establish billiard association in Montenegro (98.7%), that that change would provide more players' participation at tournaments (100%), that popularity of this sport would increase (97.4%), and that those players who are the best may become professionals (91%). Based on these results, it may be concluded that existence of an umbrella organization would mean a lot for improvement of this sport, especially in the segment of education and support for younger players.

This research results are important for theory and for practice. Theoretical significance is in all collected data which weren't available, now are in one place, and they may be base for improvement monitoring in future, after potential establishment of billiard association. Practical significance of this research is in ability to use collected data for improvement of current public relation to this sport, which is proven at high level and has its' developing potential. Also, collected data may help people who are in process of establishment of billiard association to based on them, have insight in attitudes of players and plan their future activities.

At the end, Montenegrin public should know, that adjacent Bosnia and Herzegovina has world top player, Sanjin Pehlivanovic, former U17 WPA 9-Ball World Champion and former two-time junior European Pool Championships champion who winning events 8-Ball, 9-Ball, and 10-Ball (Internet Archive, 2017; European Pocket Billiard Federation, 2016; 2016; 2017), so why wouldn't Montenegro provide chance to young people to develop their potential in this sport and make future champions such as aforementioned one.

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Conflict of Interest

The authors declare that there is no conflict of interest.

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Original Scientific Paper

Differences in the Anthropological Status of Young Football Players in Relation to the Chronological age within one Calendar Year

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Abstract

The aim of this research is to determine whether there are statistically significant differences in the anthropological status of young football players in relation to chronological age within one calendar year. The sample consisted of a total of 50 male respondents, Technical School "Mihajlo Pupin" students from the municipality of Indjija. The sample was divided into two subsamples according to chronological age (born by June 30, 2005; born July 1, 2005 and later). The students are also members of the football club in which they train. By examining the morphological characteristics, data were obtained in the following variables: body height, body mass and body mass index. Static and explosive power testing data were obtained in the following variables: standing long jump, bent arm hang and running 30 meters. It was concluded that there are no statistically significant differences between young football players of different chronological ages in anthropological status. No differences were found in the total space of tested and measured variables (MANOVA) nor in the individual space (ANOVA). The research included two anthropological spaces, morphological and motor, and none of them showed differences between young football players of different chronological ages. Out of a total of six measured and tested variables, the only variable that was on the border of the difference between the groups was running 30 meters, which may indicate that different explosive power of the lower extremities between the two tested groups, in this case, in favor of the younger group.

Keywords: *Motor Skills, Morphological characteristics, Soccer, Comparison, Male gender*

Uvod

Razvoj čovjeka je složen i diskontinuiran proces koji se odvija pod uticajem spoljnjih i unutrašnjih faktora. Genetski potencijal, geografsko područje, rasa, pol, socijalni status, ishrana, životne navike i patološka stanja mogu uticati na dinamiku i kvalitet razvoja čovjeka (Erlich, 2000; Georgiev & Gontarev, 2019). Sa ontogenetskog aspekta čovjek je pod jakim uticajima zakonitosti prirode koje determinišu redosled, pravac i smjer razvoja uz tri fundamentalne komponente: diferenciranja tkiva, funkcionalno sazrijevanje i rast (Ugarković, 2001). Tokom vremena čovjek prolazi razvojne etape koje su antropološki jasno definisane, a trajanje svake određeno je vremenskim intervalom, mada postoje individualne razlike u brzini razvoja (Banjevic, 2019). Sportisti, trenirani u određenom režimu, posjeduju značajno različite karakteristike u odnosu na prosječne ljude (Masanovic, 2019).

Zrelost mladih sportista se može posmatrati hronološki i biološki. Hronološka zrelost je posmatrani period od rođenja do zadate vremenske odrednice (Prieto, 2005). Životna dob posmatrana hronološki podrazumijeva upotrebu astroloških parametara kako bi se odredila starost. Takav pristup jednostrano posmatra razvoj čovjeka, ne uzimajući u obzir brojne uticaje koji mogu dovesti u zabludu kada se sagledava biološki potencijal organskih sistema. Napredak u razvoju je zasnovan na fiziološkim karakteristikama ljudskog organizma, nivo biološke i hronološke zrelosti često može da se ne poklapa, posebno u periodu prije adolescencije i adolescencije (Frank, 1979). Biološka zrelost je definisana po

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modelu funkcionalnog, morfološkog, anatomskog i biohemijskog statusa organizma koji je karakterističan za određeno hronološko razdoblje, a disbalans biološke i hronološke zrelosti sagledan je u brzini promjena koje karakterišu razvojne etape. Neke osobe brže sazrijevaju i ranije dosežu biološke parametre koji su karakteristični za populaciju koja je na višem hronološkom razdoblju. U literaturi se navode tri grupe biološke zrelosti u odnosu na hronološku starost: mladi koji kasne u biološkom razvoju za svojim vršnjacima, oni koji sazrijevaju u skladu sa hronološkom odrednicom i grupa u kojoj su osobe dosegle viši nivo biološke zrelosti (akceleranti) u odnosu na hronološki uzrast (Avsiyevich, 2016). Razlike bioloških parametara mogu biti latentne, ali sa morfološkog i antropometrijskog aspekta jasno se uočavaju među grupama. U funkcionalnim obilježjima primjetan je veći prirast snage kod akceleranata kao posledica višeg stepena sazrijevanja muskulature što se odražava na bolje performanse motoričkih sposobnosti i superiornost u sportskom izvođenju (Avsiyevich, 2016). U primarne somatske indikatore na osnovu kojih se vrši biološka procjena zrelosti kategorišu se mjere osifikacije hrskavice, primarno i sekundarno polno sazrijevanje, zubna zrelost, morfološki status, biohemijski i hormonski markeri (Beunen et al., 2006). Procjena biološke zrelosti mladih sportista važan je segment dijagnostike, posmatrani fenomen individualnog razvoja čovjeka predstavlja potrebu da se definišu i utvrde biološke osnove svakog pojedinca. Uspješnost u sportskoj aktivnosti u većini sportova dominantno je povezana sa morfološkom strukturom, antropometrijskim karakteristikama i motoričkim sposobnostima (Jakovljević et al., 2016). Razlike u biološkom potencijalu unutar iste generacije sportista se upravo u tim pojedinostima diferenciraju što može biti diskriminatorni faktor za postizanje sportskih rezultata. Kada se utvrde veća odstupanja biološke zrelosti od hronološkog standarda bioloških markera potrebno je primijeniti adekvatan pristup u trenažnom radu. Akceleracija trenutno doprinosi boljim sportskim rezultatima u mlađim selekcijama, ali zbog dosezanja maksimalnog razvojnog nivoa seniorski učinak je doveden u pitanje. Treneri često selektiraju igrače koji su rođeni u prvoj polovini godine ili koji su biološki zreliji kako bi ostvarili uspjeh na takmičenju (Müller, 2017). Nerijetko se dešava da sportisti koji se razvijaju u skladu sa svojim uzrastom ili oni koji kasne u razvoju, u seniorskom uzrastu ostvare bolji sportski rezultat u odnosu na one koji su taj uspjeh imali na takmičenjima mlađih kategorija (Malina, 2010). Nivo biološke zrelosti sportista prilikom selekcije, posebno tokom trenažnog procesa treba uzeti u obzir i uvažavati pri donošenju odluka. Takođe, određivanje pozicije u timu kod ekipnih sportova koje se zasniva na trenutnim tjelesnim karakteristikama i fizičkim sposobnostima, može da dovede do rane specijalizacije čime se jednostrano razvijaju vještine (Baker, Cobley, & Fraser-Thomas, 2009). Sportisti čija je biološka zrelost u skladu sa hronološkim uzrastom, posebno u periodu puberteta, mogu doživjeti emocionalne krize, anksiozna stanja koja su izazvana neuspjehom na takmičenjima sa drugim ekipama u kojima su vršnjaci, biološki zreliji, ostvarili bolji uspjeh (Horn, 2015).

Cilj istraživanja bio je da se utvrdi da li postoje statistički značajne razlike u antropološkom statusu mladih fudbalera u odnosu na hronološku starost unutar jedne kalendarske godine.

Metod

Uzorak ispitanika čini 50 ispitanika muškog pola, podijeljenih u dva subuzorka po 25 ispitanika. Svi ispitanici rođeni su 2005. godine, međutim prva grupa rođena je do 30. juna, dok je druga rođena 1. jula i kasnije. Svi ispitanici su učenici Tehničke škole „Mihajlo Pupin" sa područja grada Inđije. Učenici škole ujedno su i članovi fudbalskog kluba u kome treniraju. U istraživanje su uključeni samo oni učenici čiji su roditelji dali pismenu saglasnost.

Antropometrijsko istraživanje sprovedeno je uz poštovanje osnovnih pravila i principa vezanih za izbor mjernih instrumenata i tehnike mjerenja koji su standardizovani prema uputstvima Internacionalnog Biološkog Programa. Za potrebe ovog istraživanja izmjerene su dvije antropometrijske mjere, visina tijela (cm) i težina tijela (kg), dok je na osnovu njih, a pomoću standardne formule izračunat indeks tjelesne mase (kg/m2). Baterija testova za procjenu motoričkih sposobnosti obuhvatila je tri testa eksplozivne i statičke snage: skok u dalj iz mjesta (cm), izdržaj u zgibu (sek.) i sprint na 30 metara (sek.). Mjerenja su obavljena u fiskulturnoj sali Tehničke škole „Mihajlo Pupin" u Inđiji. Prije početka mjerenja i testiranja ispitanicima je bio detaljno objašnjen protokol testiranja.

Za svaku varijablu, od osnovnih deskriptivnih statističkih pokazatelja prikazani su aritmetička sredina i standardna devijacija. Primjenom univarijatne (ANOVA) i multivarijatne analize varijanese (MANOVA) utvrđivane su razlike u pojedinačnom i ukupnom prostoru mjerenih i testiranih varijabli između dvije grupe ispitanika. Za sve analize nivo statističke značajnosti je postavljen na r<0,05. Obrada podataka izvršena je statističkim paketom SPSS 20.0 (Chicago, IL, USA).

Rezultati

U tabeli 1. prikazane su razlike morfoloških karakteristika između mladih fudbalera u odnosu na hronološku starost unutar jedne kalendarske godine.

Tabela 1. Razlike u morfološkim karakteristikama između mladih fudbalera različite hronološke dobi (n=25)

Varijabla	Grupa	AS	SD	f	p
Tjelesna visina (cm)	Rođeni do 30. juna	178.80	8.40	0.04	0.820
	Rođeni 1. jula i kasnije	179.36	9.75		
Tjelesna masa (kg)	Rođeni do 30. juna	72.96	11.90	0.06	0.790
	Rođeni 1. jula i kasnije	72.00	14.54		
BMI (kg/m^2)	Rođeni do 30. juna	22.77	3.09	0.17	0.670
	Rođeni 1. jula i kasnije	22.35	3.97		
	F=0,85		**P=0,96**		

Legenda: AS - aritmetička sredina, SD - standardna devijacija, f - vrijednost f-testa univarijatne analize varijanse, p - nivo značajnosti univarijantne analize, F - vrijednost f-testa multivarijatne analize varijanse, P - nivo značajnosti multivarijatne analize varijanse.

Multivarijatnom analizom varijanse nijesu ustanovljene statistički značajne razlike u ukupnom prostoru morfoloških karakteristika između dvije grupe mladih fudbalera (F=0,85; P=0,96). Univarijatnom analizom varijanse nijesu utvrđene razlike ni u pojedinačnom prostoru.

U tabeli 2. prikazane su razlike motoričkih sposobnosti između mladih fudbalera u odnosu na hronološku starost unutar jedne kalendarske godine.

Multivarijatnom analizom varijanse nijesu ustanovljene statistički značajne razlike u ukupnom prostoru motoričkih sposobnosti između dvije grupe mladih fudbalera (F=1,58; P=0,20). Univarijatnom analizom varijanse nijesu utvrđene razlike ni u pojedinačnom prostoru varijabli.

Tabela 2. Razlike u motoričkim sposobnostima između mladih fudbalera različite hronološke dobi (n=25)

Varijabla	Grupa	AS	SD	f	p
Sprint na 30m (s)	Rođeni do 30. juna	4.87	0.27	3.59	0.060
	Rođeni 1. jula i kasnije	5.02	0.26		
Skok u dalj iz mjesta (cm)	Rođeni do 30. juna	197.08	14.03	0.73	0.390
	Rođeni 1. jula i kasnije	192.88	20.14		
Izdržaj u zgibu (s)	Rođeni do 30. juna	60.16	9.33	0.02	0.870
	Rođeni 1. jula i kasnije	60.60	9.99		
	F=1,58		**P=0,20**		

Legenda: AS - aritmetička sredina, SD - standardna devijacija, f - vrijednost f-testa univarijatne analize varijanse, p - nivo značajnosti univarijantne analize, F - vrijednost f-testa multivarijatne analize varijanse, P - nivo značajnosti multivarijatne analize varijanse.

Diskusija

Ovo istraživanje sprovedeno je u cilju analize antropološkog statusa dvije grupe mladih fudbalera rođenih 2005. godine. Analizom morfoloških karakteristika i motoričkih sposobnosti mladih fudbalera utvrđeno je da ne postoje statistički značajne razlike u odnosu na hronološku starost unutar jedne kalendarske godine. Naime, utvrđeno je da nema statistički značajnih razlika niti u ukupnom (MANOVA), niti u pojedinačnom prostoru varijabli (ANOVA). Slični rezultati su uočeni kod istraživanja koja su ranije sprovedena (Javorac, Smajić, Molnar, & Barašić-Huba, 2015; Popović, Molnar, & Mašanović, 2010). Uzrast ispitanika je približne decimalne vrijednosti (unutar jedne godine), što je isto jedan od razloga koji mogu da opravdaju dobijene rezultate ovog istraživanja na uzorku mladih fudbalera. Za razliku od ovog istraživanja Carling et al. (2009) su utvrdili značajne razlike u tjelesnoj visini i daljini skoka u dalj u korist fudbalera rođenih u prvom dijelu godine. Takođe, Bidaurazzaga-Letona et al. (2014) su došli do djelimično različitih zaključaka gdje su utvrdili da fudbaleri rođeni u prvom kvartalu godine postižu bolje rezultate u maksimalnoj brzini trčanja, dok razliku u antropometrijskim karakteristikama nijesu pronašli. To je veoma važno istaći kako bi treneri bili svjesni efekta relativne dobi, kako sportisti rođeni u prvom dijelu godine mogu imati prednosti i u fudbalu, i generalno u sportu. Uz dodatnu edukaciju trenera, potrebno je promijeniti fiziologiju fudbalskih klubova gdje je najčešće bitan rezultat a ne gleda se pojedinačan razvoj mladih igrača.

Na osnovu rezultata se vidi da je od ukupno šest mjerenih i testiranih varijabli (po tri iz svakog antropološkog prostora) jedino varijabla trčanje 30 metara bila na granici statistički značajne razlike između grupa, što može da ukazuje na različitu eksplozivnu snagu donjih ekstremiteta između dvije testirane starosne grupe, u ovom slučaju, u korist mlađe grupe, što opet govori da pola godine nije dovoljan period da se jave razlike.

S obzirom na prethodna istaživanja pomenuta u ovom radu, čiji su rezultati suprotni, nepostojanje statistički značajnih razlika autor može obrazložiti kroz više faktora. Prvi od njih je relativno mali uzorak i mala razlika u hronološkoj starosti. Jedna od preporuka za buduća istraživanja mogla bi biti da uzorak bude veći radi preciznije i relevantnije analize podataka, kao i da se obuhvati više uzrasta.

Limitirajući efekat ovog istraživanja je što je transferzalnog karaktera i obuhvata samo jednu uzrasnu grupu fudbalera. Ostali faktori koji su mogli uticati na rezultate mogu se pripisati velikim standardnim devijacijama, greškama pri mjerenju kao i da ispitanici tokom testiranja nijesu iskazali svoje sposobnosti u punom potencijalu. Međutim, ovakva i slična istraživanja svakako da doprinose proširenju baze podataka antropoloških karakteristika mladih sportista. Dobro poznavanje antropoloških karakteristika i sposobnosti sportista u pojedinim etapama rasta i razvoja, može doprinijeti povećanju efikasnosti, smanjenju povreda i uspješnijoj realizaciji sportskog treninga.

Rezultati ovog istraživanja mogu biti od koristi trenerima koji rade sa mladim sportistima, mogu im ukazati na veliku važnost poznavanja uzrasnih karakteristika djece koju treniraju, a posebno onih koje se odnose na hronološku i biološku zrelost, kako bi na pravilan, stručan i bezbjedan način sproveli trenažni proces, posebno dio koji se odnosi na obim, intenzitet i opterećenje, koji moraju biti prilagođeni uzrasnim, polnim, psihološkim i svim drugima karakteristikama djece koju treniraju.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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Review Paper

The Effects Of Physical Activity On Obesity Among The Population Of Different Ages: A Systematic Review

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Abstract

The aim of this research is a systematic review of the available literature with the effects of physical activity (aerobic training, strength endurance training, etc.) on the obesity of the population of different ages. For collection of previous research on the impact of physical activity on motor fitness, the following electronic databases were searched: PubMed, SCIndeks, PEDro, J-GATE, SCIndes, DOAJ and Google Scholar. The works in the period from 2000 to 2019 were searched. The following keywords were used in the database search: exercise, physical activity, children, adult, aerobic training, resistance training, walking. This systematic review was conducted in agreement with the PRISMA guidelines. The results of the analyzed works indicated that only 20 works met the set selection criteria. In the analyzed works were 845 respondents. In the most researches, the training program lasted 12 weeks, while the shortest program lasted only 5 weeks. Combining endurance training with aerobic training has been shown as the most effective method in the prevention and treatment of obesity.

Keywords: *BMI, Nutrition, Aerobic Training, Physical Activity, Fat Mass, Muscle Mass*

Introduction

Diet and physical activity directly affect the health status of adults and children (Strong et al., 2005; Warburton, Nicol, & Bredin, 2006; Janssen & Leblanc, 2010; Petrović - Oggiano, Damjanov, Gurinović, & Glibetič, 2010; Mitić, 2001; 2011). The generally accepted scientific definition identify physical activity as „any bodily movement produced by skeletal muscles that results in energy expenditure“ (Caspersen, Powell, & Christenson, 1985). Physical activity is one of the key determinants of energy consumption and for this reason is very important for maintaining energy balance and weight control (World Health Organization, 2000; 2016).

To determine the prevalence of obesity and obesity in the population, the most commonly used estimate of nutritional levels based on body mass index (BMI - body mass index), which represents the ratio of body weight and square height. Previous research has shown that physical inactivity is negatively associated with BMI, waist circumference, and body fat percentage in both sexes (Du et al., 2013; Kaleta, Makowiec-Dabrowska, & Jegier, 2007). Physical activity increases energy consumption and thus affects the overall daily balance of energy intake and consumption. Weight gain occurs when the energy balance is positive, i.e. when energy intake is greater than consumption. (Zdravković, Banićević, & Petrović, 2009).

In children and adolescents, the definition of obesity is more complex because the total body fat content depends on the chronological age, sex and stadium of pubertal development (Leskošek, 1971). In children, due to normal changes in BMI (body mass index), specific for age and sex, percentiles are used, which are determined by entering the calculated BMI in the growth chart of body mass index specific for the sex of the respondents. At the age of 20, the value of BMI above the 85th percentile represents overnutrition, i.e. risk of obesity, from the 95th percentile is defined as obesity. Less than 5% of childhood causes of

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obesity are secondary obesity, associated with genetic disorders, endocrine diseases, lesions of the central nervous system or iatrogenic causes (Leskošek, 1971). The most significant factors that contribute to the growing obesity epidemic are physical inactivity, sedentary lifestyle and changes in diet (Štimec, 2012). Based on the aforementioned factors that contribute to the growing obesity epidemic, it can be concluded that physical activity is one of the most important factors for the prevention and treatment of obesity (Lee, Djoussé, Sesso, Wang, & Buring, 2010; Saris, 1998).

The aim of this systematic review was to gather available information on the impact of different types of physical activity on obesity depending on age and to draw a conclusion from this which physical activity gives the best results in practice.

Methods

Inclusion criteria

For an experimental study to be included in the final analysis, it had to meet certain criteria: the participant in the research were individuals suffering from obesity; the experimental study included subjects of both sexes; respondents participate in exercise programs during which its effects were evaluated and measurements taken of the parameters of body composition; the research was published in English or Serbian. The exclusion criteria included: papers not published in English or Serbian, studies in which the effect of physical activity on body composition has not been analyzed.

Search strategy

To collect previous research on the impact of physical activity on motor fitness, the following electronic databases were searched: PubMed, SCIndeks, PEDro, J-GATE, SCIndes, DOAJ and Google Scholar. The works in the period from 2000 to 2019 were searched. The following keywords were used in the database search: exercise, physical activity, children, adult, aerobic training, resistance training, walking. The found research titles, abstracts and full texts were then read and analyzed. This systematic review was undertaken in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Moher et al., 2009).

Data extraction and selection

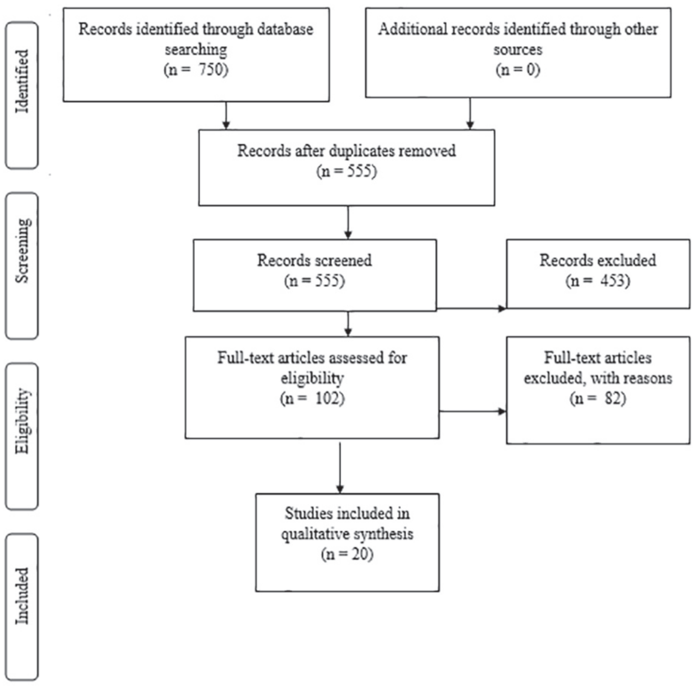
Experimental research which met the set criteria was then analyzed and presented based on the following parameters: references (the initials of the author and year of publication), the sample of participants (gender, BMI, age, overall number and subgroups of the participants), physical exercise program, the duration and frequency of exercise, research results.

Study quality and risk of bias

Risk of bias was evaluated according to the PRISMA recommendation and two independent reviewers assessed the risk of bias. When there was disagreement about the risk of bias a third reviewer checked the data and took the final decision on it.

Synthesis of results

By searching the electronic databases, 750 experimental studies were identified on the set topic. Primarily 195 studies were eliminated as duplicates, then 453 papers were rejected based on abstracts, while 82 studies were excluded because they were of the review type, or were not written in English. Applying the set criteria, the final analysis included 20 experimental studies.



GRAPH 1. Procedure of collection, analysis and elimination of found scientific research

Results

In this systematic review, overall 20 papers were analyzed. In 10 of 20 papers, the authors addressed the impact of physical activity on the treatment of obesity in children (Alberga, Farnesi, Lafleche, Legault, & Komorowski, 2013; Farris, Taylor, Williamson, & Robinson, 2011; Lee, Bacha, Hannon, Kuk, Boesch, & Arslanian, 2012; McGuigan, Tatasciore, Newton, & Pettigrew, 2009; Mendelson et al., 2015; Prado et al., 2009; Rey, Vallier, Nicol, Mercier, & Maïano, 2017; Van der Heijden et al., 2010; Wong et al., 2008; Yu et al., 2005), while the remaining 10 papers analysed the effects of physical activity in adults obesity (Dobrosielski, Patil, Schwartz, Bandeen-Roche, & Stewart, 2015; Fogelholm, Kukkonen-Harjula, Nenonen, & Pasanen, 2000; Macura & Cirkovic, 2014; Meredith-Jones, Legge, & Jones, 2009; Mladenov, 2014; Okura, Nakata, Lee, Ohkawara, & Tanaka, 2005; S. Park, J. Park, Kwon, Kim, Yoon, & H. Park, 2003; Phillips et al., 2012; Saif & Alsenany, 2015; Skryp-

Table 1. Review of analysed researchs

Study	Respondents					Applied training program			Output data	
	Gender	Age	BMI	No of respondents	Subgroups	Program duration (weeks/ days)	Duration of training (min)	Type of activity	Result	Conclusion
Alberga et al., 2013	M/F	8 - 12	/	19	EG KG	12 / 2	75	VITS	After vs before Body mass (57.6 ± 13.5 vs 59.6 ± 14.1 kg) Muscle mass (32.6 ± 6.8 vs 34.0 ± 7.0 kg)	High-intensity strength training has a positive effect on body composition, after 12 weeks of training in children aged 8 to 12 there was a decrease in body weight and BMI.
Farris et al., 2011	M/F	6 - 12	>95%	25	EG	12 / 2	60	AT + TI	Body composition was better after the program than before the program.	Aerobic training with endurance training has a positive effect on body composition.
Lee et al., 2012	M	12-18	>95%	43	KG EG1 EG2	12 / 3	60	AT TI	After the program, both experiments had a reduced body weight (aerobic training -0.004 ± 0.8 kg; endurance training -0.6 ± 0.8 kg), while body weight in the control group increased.	The obtained results indicate that both aerobic and endurance training have a positive effect on weight loss in boys aged 12 to 18 years. However, it should be noted that endurance training gives better results than aerobic training.
McGuigan, et al., 2009	M/F	9,7	25,6 ± 3,1	48		8 / 3	X	TS	After the program, the average value of the percentage of fat in the subjects decreased by 2.6%.	Strength training has a positive effect on reducing the percentage of fat in adolescents from 7 to 12 years.
Mendelson et al., 2015	M/F	14.5±1.5	≥95%	40	EG KG	12 / 2	X	AT TI	After the program in the experimental group, body weight, abdominal and visceral fat remained unchanged, while body fat decreased	The obtained results indicate that aerobic training with endurance training significantly affects only the reduction of fat, while body weight remains unchanged.
Prado et al., 2009	M/F	10±0,2	30+/-1	38	KG EG	X	X	X	After the program, all obese children reduced their body weight (p <0.05).	Based on the obtained results, it was shown that there is no statistically significant difference in children whether only diet or both diet and training were applied.

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Study	Respondents					Applied training program			Output data	
	Gender	Age	BMI	No of respondents	Subgroups	Program duration (weeks/ days)	Duration of training (min)	Type of activity	Result	Conclusion
Rey et al., 2017	M/F	14 - 15	>95%	24	EG	5	X	VIT	After the applied program, there was a reduction in body weight, body fat and BMI.	Based on the obtained results, it can be concluded that high-intensity interval training has a positive effect on weight reduction, body fat and BMI in subjects aged 14 to 15 years of both sexes.
Van der Heijden et al., 2010	M/F	15,6±0,4	33,7 ± 1,1	15	/	12 / 4	30	AT	After 12 weeks of the program, body weight, fat percentage, BMI, fat mass decreased.	The obtained results indicate that aerobic training is suitable and the correction of the body composition of obese children.
Wong et al., 2008	M/F	13-14	30.6 ± 2.1 31.8 ± 4.4	24	EG KG	12 / 2	45-60	AT + TI	After 12 weeks, the experimental group reduced their body weight, fat percentage and BMI, while muscle mass increased.	Aerobic training in combination with endurance training has a positive effect on the body composition of adolescents aged 13 to 14 years.
Yu et al., 2005	M/F	8 - 11	25,5 24,7	82	EG KG	6 / 3	75	AT TS TA	After six weeks of the program, the subjects of the experimental group had a greater increase in muscle mass compared to the control group (+0.8 kg [2.4%] vs. +0.3 kg [1.0%], p <0.05). Also, the percentage of fat in the experimental group decreased by 0.7%, while in the control group by only 0.2%.	Research has shown that strength training in children aged 8 to 11 has a better effect than dieting. After six weeks, the subjects of the experimental group had better results when it comes to the proportion of fat and muscle mass compared to the subjects of the control group.

M – male; F – female; X – no data specified; EG – experimental group; KG - control group; AT – aerobic training; TS - strength training; TA - agility training; ANT - anaerobic training; TI - endurance training; VIT - high-intensity interval training; CT - combined training; VITS - high-intensity strength training.

Table 2. Review of analysed researchs for adults

Study	Respondents					Applied training program			Output data	
	Gender	Age	BMI	No of respondents	Subgroups	Program duration (weeks/ days)	Duration of training (min)	Type of activity	Result	Conclusion
Dobrosielski et al., 2015	M/F	≥60	33.8	25	/	X	X	AT	After the applied program, there was a reduction in body weight, fat percentage and BMI.	Based on the obtained results, it can be concluded that aerobic training has a positive effect on the body composition of adults.
Fogelholm et al., 2000	F	40 - 45	30-45	80	KG EG1 EG2	40	X	Walking	After a period of 40 weeks, the control group increased its average body weight, while this value decreased in both experimental groups, with the group with lower intensity achieving a slightly better result.	The obtained results show that walking as a type of physical activity is effective for losing excess weight.
Macura & Cirkovic, 2014	F	34,2	/	9	EG	8 / 3	60-75	AT ANT	In the assessment of body composition, statistical significance was obtained in body mass, body fat, muscle mass, waist circumference and thigh circumference.	Based on the obtained results, it has been shown that the continuous training method has a positive effect on the body composition of women.
Meredith-Jones et al., 2009	F	59 ± 9	33 ± 5	18	/	12 / 3	60	KT	The results showed that when it comes to body composition, such a program had a statistically significant effect only on the ratio of waist and hips.	Based on the obtained results, it can be concluded that such a high-intensity program does not have a significant impact on reducing fat and body weight in obese people.
Mladenov, 2014	F	31,33	25.29	9	/	12 / 3	60	AT	The average value of BMI (kg / m2) at the initial measurement was 25.29 kg / m2 while at the final measurement it was 23.97 kg / m2.	Based on the obtained results, it can be concluded that physical aerobic activity with diet has a much better effect on body composition than diet alone.
Okura et al., 2005	F	21 - 66	>25	209	EK KG	14	X	AT	A statistically significant difference appeared only in the respondents who were in the group that applied a program of aerobic exercises in addition to the diet. Significantly decreased body weight, body mass index, fat percentage, total adipose tissue mass.	Based on the obtained results, it can be concluded that physical aerobic activity with diet has a much better effect on body composition than diet alone.

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Study	Respondents					Applied training program			Output data	
	Gender	Age	BMI	No of respondents	Subgroups	Program duration (weeks/ days)	Duration of training (min)	Type of activity	Result	Conclusion
Park et al., 2003	F	40 - 45	25,56±0,86 25,3 ± 1,74 25,8 ± 1,43	30	KG EG1 EG2	24 / 6	60	AT AT + TI	After 24 weeks, the following results were obtained Body mass: KG +0.6; AT -4.7; CT -6.4 Fat percentage: KG +2.3; AT -9.2; CT -10.3 Muscle mass: KG -0.4; AT +0.9; CT + 5.6	Based on the obtained results, it can be concluded that both combined and aerobic training have a positive effect on body composition. There is no statistically significant difference between the groups that applied aerobic and combined training, but it can be noticed that the subjects who applied combined training achieved better results than the subjects who applied only aerobic training.
Phillips et al. 2012	F	65±2,6	33	23	EG KG	12 / 3	X	TS	After the applied program in the experimental group, there was a decrease in body weight, the percentage of BMI fat.	Based on the obtained data, it can be concluded that strength training has a positive effect on body composition.
Saif & Alsenany, 2015	M/F	18 - 25	36.45±3.36	40	A B	12 / 3		AT ANT	After the program was applied, both groups reduced their average BMI.	The obtained results showed that both aerobic and anserobic activity for adults aged 18 to 25 years have a positive effect on BMI reduction.
Skrypnik et al., 2015	F	18 - 56	35,23 ± 3,9 34,9 ± 3,8	44	A B	12 / 3	60	TI TI + TS	In both groups, there was a significant reduction in body weight, body mass index and adipose tissue after the program. However, in group B there was a significant increase in muscle mass and the mass of non-fat components.	Both physical activities have a positive effect on the body composition of women, but it was also concluded that strength training has a positive effect on increasing muscle mass and non-fat components.

M – male; F – female; X – no data specified; EG – experimental group; KG - control group; AT – aerobic training; TS - strength training; TA - agility training; ANT - anaerobic training; TI - endurance training; VIT - high-intensity interval training; CT - combined training; VITS - high-intensity strength training.

nik et al., 2015). There were 845 participants overall, where 358 were children and 487 were adults. Lee et al., (2012) was only study with male participants, boys aged 12 to 18 years. In 11 of the 20 papers, subjects were of both sexes (Alberga et al., 2013; Dobrosielski et al., 2015; Farris et al., 2011; McGuigan et al., 2009; Mendelson et al., 2015; Prado et al., 2009; Rey et al., 2017; Saif & Alsenany, 2015; Van der Heijden et al., 2010; Wong et al., 2008; Yu et al., 2005). In the eight papers, the participants were only women (Fogelholm et al., 2000; Macura & Cirkovic, 2014; Meredith-Jones et al., 2009; Mladenov, 2014; Okura et al., 2005; Park et. al., 2003; Phillips et al., 2012; Skrypnik et al., 2015), while in remaining one paper, the participant were only men (Lee et al., 2012). In most papers, the training program lasted 12 weeks (Alberga et al., 2013; Farris et al., 2011; Lee et al., 2012; Mendelson et al., 2015; Meredith-Jones et al., 2009; Mladenov, 2014; Phillips et al., 2012; Saif & Alsenany, 2015; Skrypnik et al., 2015; Van der Heijden et al., 2010; Wong et al., 2008), the shortest program lasted only five weeks (Rey et al., 2017).

As far as physical activities are concerned, aerobic training program was the most used training program in this systematic review, 10 of 20 papers. There were obese children in four studies (Lee et al., 2012; Mendelson et al., 2015; Van det Heijden et al., 2010; Yu et al., 2005) an adults in six (Dobrosielski et al., 2015; Macura & Cirkovic, 2014; Mladenov, 2014; Okura et al., 2005; Park et al., 2003; Saif & Alsenany, 2015). In most of already mentioned papers, BMI was decreased the most (Dobrosielski et al., 2015; Mladenov, 2014; Okura et al., 2005; Saif & Alsenany, 2015; Van det Heijden et al., 2010). Second most used training program was the combination of aerobic training and endurance training program. Farris et al., (2011) and Wong et al., (2008) had children population, while Park et al., (2003) had adults and this training program decreased body weight and BMI, which resulted increasing muscle mass, in both children and adults. Only one paper (Alberga et al., 2013) applied high intensity interval training, which resulted reduction in body mass, muscle mass and BMI, while Fogelholm et al. (2000) applied walking program, which resulted better results in treatment of obesity in diet group. Macura & Cirkovic (2014) applied anaerobic training program, resulting correction of all measured body composition variables (body fat, body mass, muscle mass, waist and thigh circumference), while McGuigan, et al. (2009) used strength training as physical exercise program, which resulted body fat reduction.

Discusion

This analysis of previous experimental work provides an opportunity to see which types of physical activities are used to combat obesity. Certainly the most common type of program that was applied was the aerobic type of training. However, it is not the most effective type. Aerobic training in combination with endurance training proved to be the most effective type of training (Farris et al., 2011; Park et al., 2003; Meredith-Jones et al., 2009). It's a training method that lasts from 20 to 60 minutes, with low-intensity exercises and a heart rate zone of 50 to 75%. During such training, fats are mostly used as energy sources, which affects the reduction of body fat. However, such training can lead to a decrease in muscle mass, so it is combined with endurance training, which affects the increase in muscle mass (Rey et al., 2017).

The second most effective in combating obesity would be aerobic training alone. Aerobic physical activity is of low intensity and longer duration and as such provides energy without the appearance of lactic acids. Although this type had positive effect on weight loss, fat percentage and BMI, this phenomenon can lead to a decrease in muscle mass or a significantly smaller increase in muscle mass compared to other physical activity (Park et al., 2003).

And as the least effective types of training we are left with the remaining types such as strength training, endurance training, anaerobic training, high-intensity interval training. In all these studies, it was shown that they had a positive effect on reducing the fat percentage. However, these types of trainings are short, so the mechanism of using fat as an energy source is not activated, and primarily for these trainings is an increase in muscle mass, which is the reason for reducing the percentage of fat in the body. More precisely, the mass of adipose tissue itself changes very little, almost not at all (Meredith-Jones et al., 2009; Saif & Alsenany, 2015; Skrypnik et al., 2015). Based on the obtained results of previous research, it can be concluded that physical activities have a positive effect on obesity regardless of the age of the respondents. All types of training are used for both children and adults.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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Short Report

Cost of Healthcare is Skyrocketing in the USA: A Broken Medicare Must be Reformed

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Abstract

This short report discusses the current cost of the healthcare system in the United States. The affordability problems in the United States stem directly from insufficient competition of drug prices from the Medicare system. This report provided evidence about how the proposed Medicare reforms are skewed and would unlikely end the blank-check approach to pricing of Medicare that drives our healthcare spending and affordability problems.

Keywords: *Health Insurance, Inflation, Affordability*

Introduction

American's healthcare expenses have reached the point where it's very expensive to afford (Kaiser Family Foundation, 2019) and can be financially devastating (Himmelstein, Lawless, Thorne, Foohey, & Woolhandler, 2019).

Methods

Dusetzina and colleagues (Dusetzina, Muluneh, Keating, & Huskamp, 2020) draw attention to current drug prices from Medicare Part D, which aim to provide critical financial relief for patients and current and future taxpayers.

Results

According to their findings, current law failed to deliver promised savings, which is exact reason to restructure the Medicare Part D's benefits under the new Biden Administration. Unfortunately, current proposals (i.e., US senator proposal or congress proposal) (Dusetzina, Keating, & Huskamp, 2019) are skewed and could yield the opposite result. Specifically, proposed reforms would have similar effects on out-of-pocket spending for 90% of Part D patients (Dusetzina, Keating, & Huskamp, 2019). As a result of cost sharing, ironically, reforms could at first drive premiums drastically higher, which to the detriment of the vast majority.

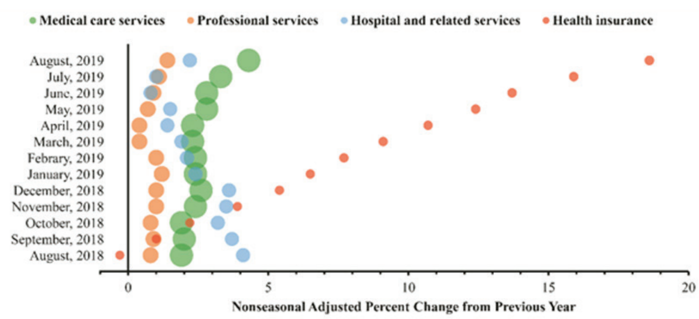

FIGURE 1. The Healthcare Annul Increase in U.S. City Average, All Urban Consumers. The medical care services index comprises the professional services, hospital and related services, and health insurance categories of the Consume Price Index (CPI). Bubble size represents relative importance among all CPI components. Data were retrieved from U.S. Bureau of Labor Statistics monthly CPI reports.

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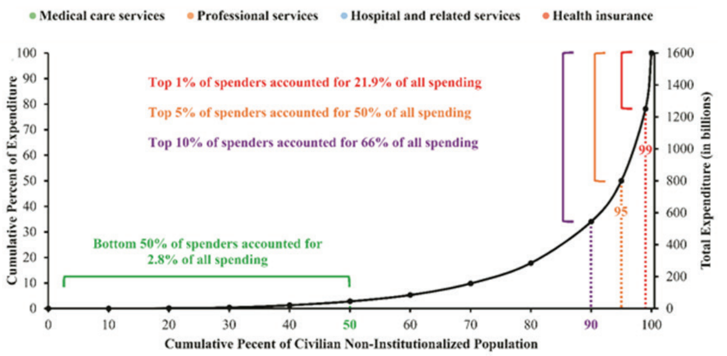

FIGURE 2. Distrubtion of U.S. Heathcare Spending by Intensity of Use. Healthcare expenditures include all sources of payment for medical care, including private insurance payments, Medicare, Medicaid, out-of-pocket spending, and other sources. Data were retrieved from Medical Expenditure Panel Survey, 2016.

Discussion

The cost of health insurance is outpacing inflation, increasing 18.6% from August 2018 to August 2019 (Figure 1) and this trend has been steady to date (U.S. Bureau of Labor Statistics, 2021). Any policy redesign should target at a reasonable price control mechanism to address the notable healthcare expenses resulting from top 5% population (Figure 2) (Mitchell, 2019) whilst offer a betterment of affordable healthcare price of many. Otherwise, these reform efforts are at a risk of pooling healthcare expenses to individual patients and non-Medicare taxpayers that increases more financial burdens for American citizens.

Acknowledgements
There are no acknowledgements.

Conflict of Interest
The authors declare that there is no conflict of interest.

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Dusetzina, S.B., Keating, N.L., & Huskamp, H.A. (2019). Proposals to redesign Medicare Part D—easing the burden of rising drug prices. *New England Journal of Medicine, 381*(15), 1401-1404.

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Mitchell, E. (2019). *Concentration of Health Expenditures and Selected Characteristics of High Spenders, U.S. Civilian Noninstitutionalized Population, 2016*. Rockville, MD: Agency for Healthcare Research and Quality.

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Revised October 2017

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Original Scientific Paper

Diet and Body Composition of Female Athletes

Svetlana Nepocatych[1], Gytis Balilionis[1], Eric K. O'Neal[2]

[1]Elon University, Department of Exercise Science1, Elon, NC 27215
[2]University of North Alabama, Department of Health, Physical Education and Recreation, Florence, AL 35632

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S. Nepocatych

Elon University

Department of Exercise Science

100 Campus Dr.

2525 CB

Elon, NC 27244

United States

E-mail: snepocatych@elon.edu

Word count: 2,946

Word count: 4259

Abstract word count: 211

Number of Tables: 3

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Title should be short and informative and the recommended length is no more than 20 words. The title should be in Title Case, written in uppercase and lowercase letters (initial uppercase for all words except articles, conjunctions, short prepositions no longer than four letters etc.) so that first letters of the words in the title are capitalized. Exceptions are words like: "and", "or", "between" etc. The word following a colon (:) or a hyphen (-) in the title is always capitalized.

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Short running title should not exceed 50 characters including spaces.

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The form of an author's name is first name, middle initial(s), and last name. In one line list all authors with full names separated by a comma (and space). Avoid any abbreviations of academic or professional titles. If authors belong to different institutions, following a family name of the author there should be a number in superscript designating affiliation.

2.1.5. Affiliations

Affiliation consists of the name of an institution, department, city, country/territory (in this order) to which the author(s) belong and to which the presented / submitted work should be attributed. List all affiliations (each in a separate line) in the order corresponding to the list of authors. Affiliations must be written in English, so carefully check the official English translation of the names of institutions and departments.

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If an author belongs to more than one institution, all corresponding superscript digits, separated with a comma with no space in between, should be present behind the family name of this author.

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Whenever possible expand your authors' affiliations with departments, or some other, specific and lower levels of organization.

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Corresponding author's name with full postal address in English and e-mail address should appear, after the affiliations. It is preferred that submitted address is institutional and not private. Corresponding author's name should include only initials of the first and middle names separated by a full stop (and a space) and the last name. Postal address should be written in the following line in sentence case. Parts of the address should be separated by a comma instead of a line break. E-mail (if possible) should be placed in the line following the postal address. Author should clearly state whether or not the e-mail should be published.

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Abstract

Results of the analysis of

Key words: *spatial memory, blind, transfer of learning, feedback*

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One work by one author
- ✓ In one study (Reilly, 1997), soccer players
- ✓ In the study by Reilly (1997), soccer players
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Works by two authors
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- ✓ In one study (Duffield & Marino, 2007), soccer players
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- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

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Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology, 100*(6), 727-735. doi: 10.1007/s00421-007-0468-x

Krustrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise, 35*(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

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Conference paper:

Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf

Encyclopedia entry (print, with author):

Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

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Book:
Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:
Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from Routledge Handbook of Applied Sport Psychology ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:
Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

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Tables should be presented as standard MS Word tables.

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Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

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- ✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

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- ✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. [a, b, c]), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript [a].

- ✓ [a]One participant was diagnosed with heat illness and n = 19.[b]n =20.

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- ✓ *P<0.05,†p<0.01.

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- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
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2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See* example:

- ✓ Figure 1
    - × figure 1
    - × Figure 1.
    - ✓ ….exhibit greater variance than the year before (Figure 2). Therefore…
    - ✓ ….as shown in Figures 1 and 3. (citing more figures at once)
    - ✓ ….result has shown (Figures 1-3) that... (citing more figures at once)
    - ✓ ….in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See* example:

- ✓ Figure 1a
- ✓ …in Figures 1a and b we can…
- ✓ …data represent (Figures 1a-d)…

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10º	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 º	× 10kg	× 12 : 2	× .056

Signs should be placed immediately preceding the relevant number.		
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age
× 45 ± 3.4	× p < 0.01	× males > 30 years of age

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. *See* example:

- ✓ First time appearing: *musculus biceps brachii*

Abbreviated: *m. biceps brachii*

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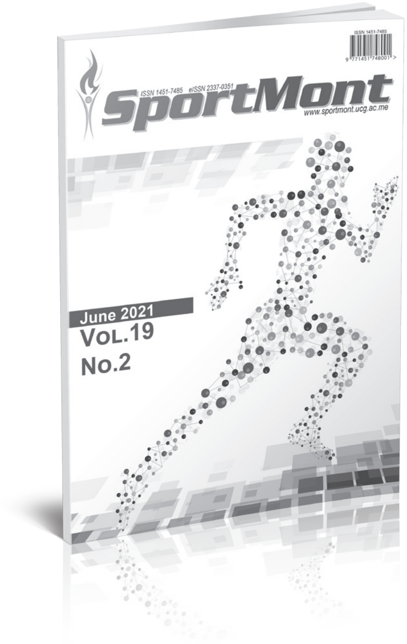

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Sport Mont Journal (SMJ) is a print (ISSN 1451-7485) and electronic scientific journal (eISSN 2337-0351) aims to present easy access to the scientific knowledge for sport-conscious individuals using contemporary methods. The purpose is to minimize the problems like the delays in publishing process of the articles or to acquire previous issues by drawing advantage from electronic medium. Hence, it provides:

- Open-access and freely accessible online;
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- Post-publication tools to indicate quality and impact;
- Community-based dialogue on articles;
- Worldwide media coverage.

SMJ is published three times a year, in February, June and October of each year. SMJ publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest.

SMJ covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

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Summer issue – June 2022

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SCIENTIFIC PROJECT

Efekti autofagije i fizičke aktivnosti na tjelesnu kompoziciju, indeks tjelesne mase, stres, zdravstveno ponašanje, kognitivne sposobnosti i društvenu uključenost starijih osoba

(Projekat finansiran od strane Ministarstva nauke, direktorijata za mlade, a koji sprovodi Fakultet za sport i fizičko vaspitanje Univerziteta Crne Gore)

Glavni cilj projekta je da se primjenom naprednih praksi utvrde efekti autofagije i fizičke aktivnosti na tjelesnu kompoziciju, indeks tjelesne mase, trenutni nivo doživljaja stresa, zdravstveno ponašanje, kognitivne sposobnosti i nivo društvene uključenosti starijih osoba.

Projekat ima i svoj radni dio koji podrazumijeva organizovanje besplatnog vježbanja starijim osobama 3 puta nedeljno u trajanju od 6 mjeseci.

Učesnici mogu biti sve osobe iznad 50 godina a prijave se primaju na mail adresu fakultetzasportnk@ucg.ac.me i telefon fakulteta +38240235207. Svi su koji žele da uzmu aktivno učešće u ovom projektu koji njima može unaprijediti život, a naučnoj zajednici Crne Gore može donijeti značajna teorijska znanja koja će se u budućnosti koristiti u praktične svrhe su dobrodošli.

Svim učesnicim će na početku biti ponuđena najsavremenija dijagnostika, koja će im pružiti uvid u vlastito zdravstveno stanje, kako fizičko tako i kada su neki psihološki parametri u pitanju. Nakon toga će učesnici biti prema sopstvenim interesovanjima podijeljeni u grupe koje će raditi prema različitim programima.

1. Prva grupa će vježbati 3 puta nedeljno 6 mjeseci u prostorijama Fakulteta za sport i fizičko vaspitanje.

2. Druga će samo primjenjivati izmijenjeni način ishrane i voditi tačnu evidenciju o stepenu poštovanja zadataka koji im se postave. Ishrana će biti takva da se napravi pauza u unošenju hrane u trajanju 16 sati između poslednjeg dnevnog obroka i prvoga obroka u sledećem danu, a sve u cilju pokretanja procesa Autofagije koji ima blagotvorno dejstvo na organiza.

3. Treća grupa će kombinovati vježbanje i izmijenjenu ishranu, tj. biće kombinacija prethodno pomenutih zadataka.

4. Četvrta grupa će biti kontrolna. Njeni članovi će proći dijagnostiku i pomoći da se utvrdi kakve su prirodne promjene u organizmu za pomenuti šestomjesečni period, odnosno da li ih ima.

Svim prijavljenim osobama, koje imaju interesovanje za to, će prije poćetka rada biti održana dva predavanja o pomenutom izmijenjenom načinu ishrane koji danas postaje sve popularniji u svijetu pa ga primjenjuju i vrhunski sportisti poput Novaka Đokovića.

Još jednom treba napomenuti da će svaka od 4 grupe na poklon dobiti najsavremeniju dijagnostiku kompletnog psihofizičkog stanja koja je inače i nedostupna i skupa.

Prijavljivanje može da počne odmah, broj učesnika za grupe koje bi vježbale u prostorijama fakulteta je ograničen.

VLADA CRNE GORE
MINISTARSTVO NAUKE

Fakultet za sport i fizičko vaspitanje
Univerzitet Crne Gore

Fakultet za sport i fizičko vaspitanje Univerziteta Crne Gore
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Montenegrin Journal
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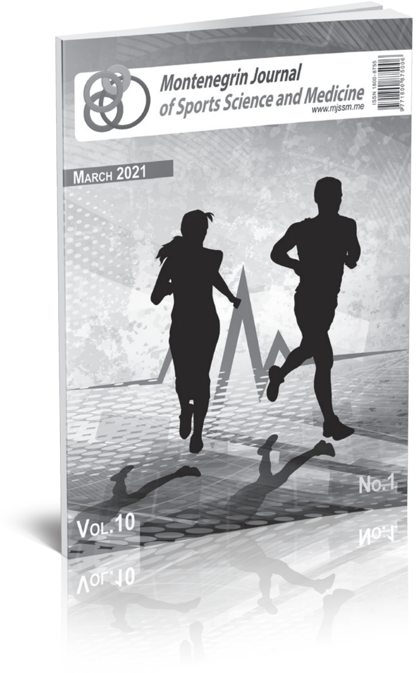

ISSN 1800-8755

CALL FOR CONTRIBUTIONS

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MJSSM covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

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MONTENEGRIN SPORTS ACADEMY

Founded in 2003 in Podgorica (Montenegro), the Montenegrin Sports Academy (MSA) is a sports scientific society dedicated to the collection, generation and dissemination of scientific knowledge at the Montenegrin level and beyond.

The Montenegrin Sports Academy (MSA) is the leading association of sports scientists at the Montenegrin level, which maintains extensive co-operation with the corresponding associations from abroad. The purpose of the MSA is the promotion of science and research, with special attention to sports science across Montenegro and beyond. Its topics include motivation, attitudes, values and responses, adaptation, performance and health aspects of people engaged in physical activity and the relation of physical activity and lifestyle to health, prevention and aging. These topics are investigated on an interdisciplinary basis and they bring together scientists from all areas of sports science, such as adapted physical activity, biochemistry, biomechanics, chronic disease and exercise, coaching and performance, doping, education, engineering and technology, environmental physiology, ethics, exercise and health, exercise, lifestyle and fitness, gender in sports, growth and development, human performance and aging, management and sports law, molecular biology and genetics, motor control and learning, muscle mechanics and neuromuscular control, muscle metabolism and hemodynamics, nutrition and exercise, overtraining, physiology, physiotherapy, rehabilitation, sports history, sports medicine, sports pedagogy, sports philosophy, sports psychology, sports sociology, training and testing.

The MSA is a non-profit organization. It supports Montenegrin institutions, such as the Ministry of Education and Sports, the Ministry of Science and the Montenegrin Olympic Committee, by offering scientific advice and assistance for carrying out coordinated national and European research projects defined by these bodies. In addition, the MSA serves as the most important Montenegrin and regional network of sports scientists from all relevant subdisciplines.

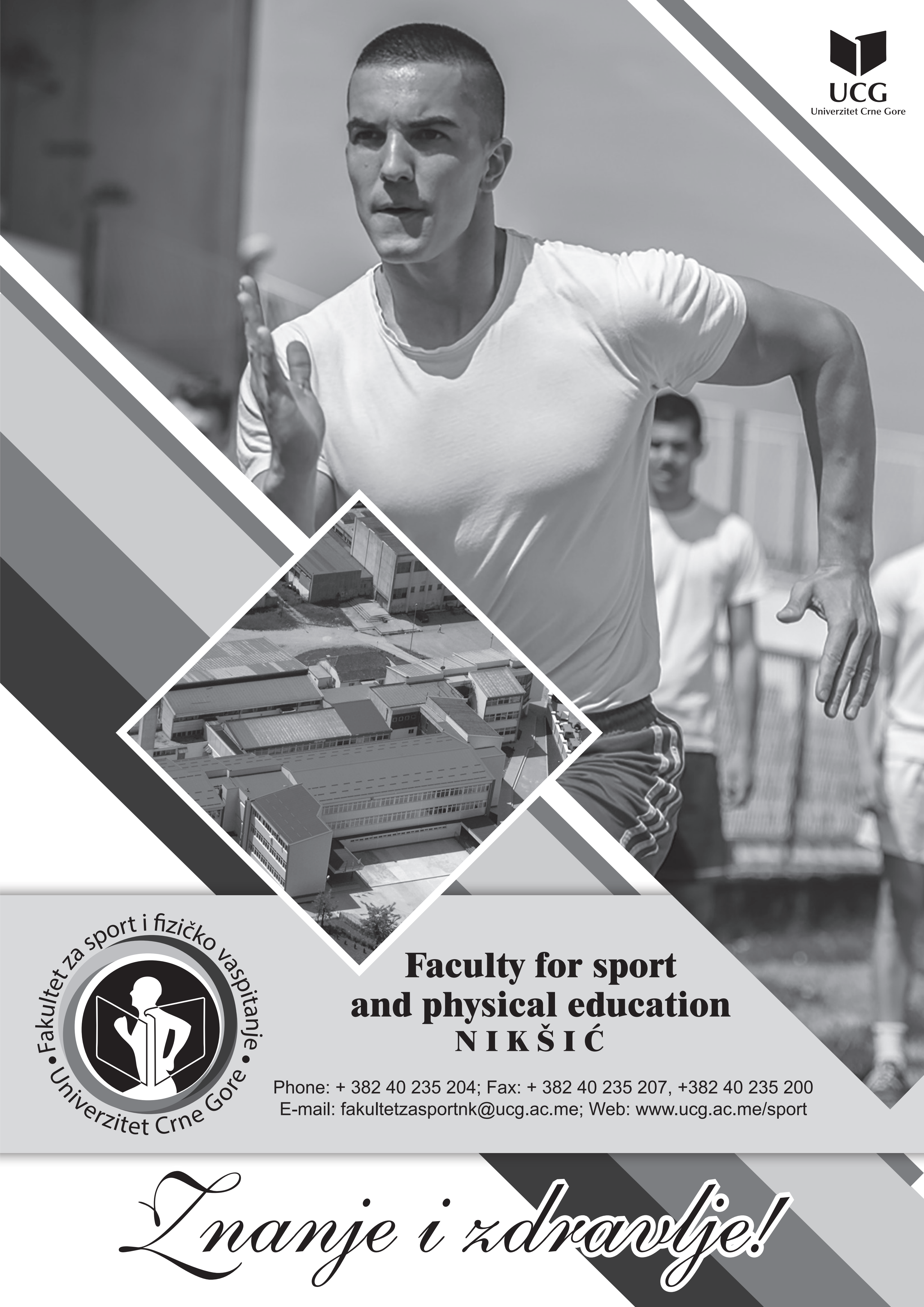
The main scientific event organized by the Montenegrin Sports Academy (MSA) is the annual conference held in the first week of April.

Annual conferences have been organized since the inauguration of the MSA in 2003. Today the MSA conference ranks among the leading sports scientific congresses in the Western Balkans. The conference comprises a range of invited lecturers, oral and poster presentations from multi- and mono-disciplinary areas, as well as various types of workshops. The MSA conference is attended by national, regional and international sports scientists with academic careers. The MSA conference now welcomes up to 200 participants from all over the world.

It is our great pleasure to announce the upcoming 19th Annual Scientific Conference of Montenegrin Sports Academy “Sport, Physical Activity and Health: Contemporary Perspectives” to be held in Dubrovnik, Croatia, from 7 to 10 April, 2022. It is planned to be once again organized by the Montenegrin Sports Academy, in cooperation with the Faculty of Sport and Physical Education, University of Montenegro and other international partner institutions (specified in the partner section).

The conference is focused on very current topics from all areas of sports science and sports medicine including physiology and sports medicine, social sciences and humanities, biomechanics and neuromuscular (see Abstract Submission page for more information).

We do believe that the topics offered to our conference participants will serve as a useful forum for the presentation of the latest research, as well as both for the theoretical and applied insight into the field of sports science and sports medicine disciplines.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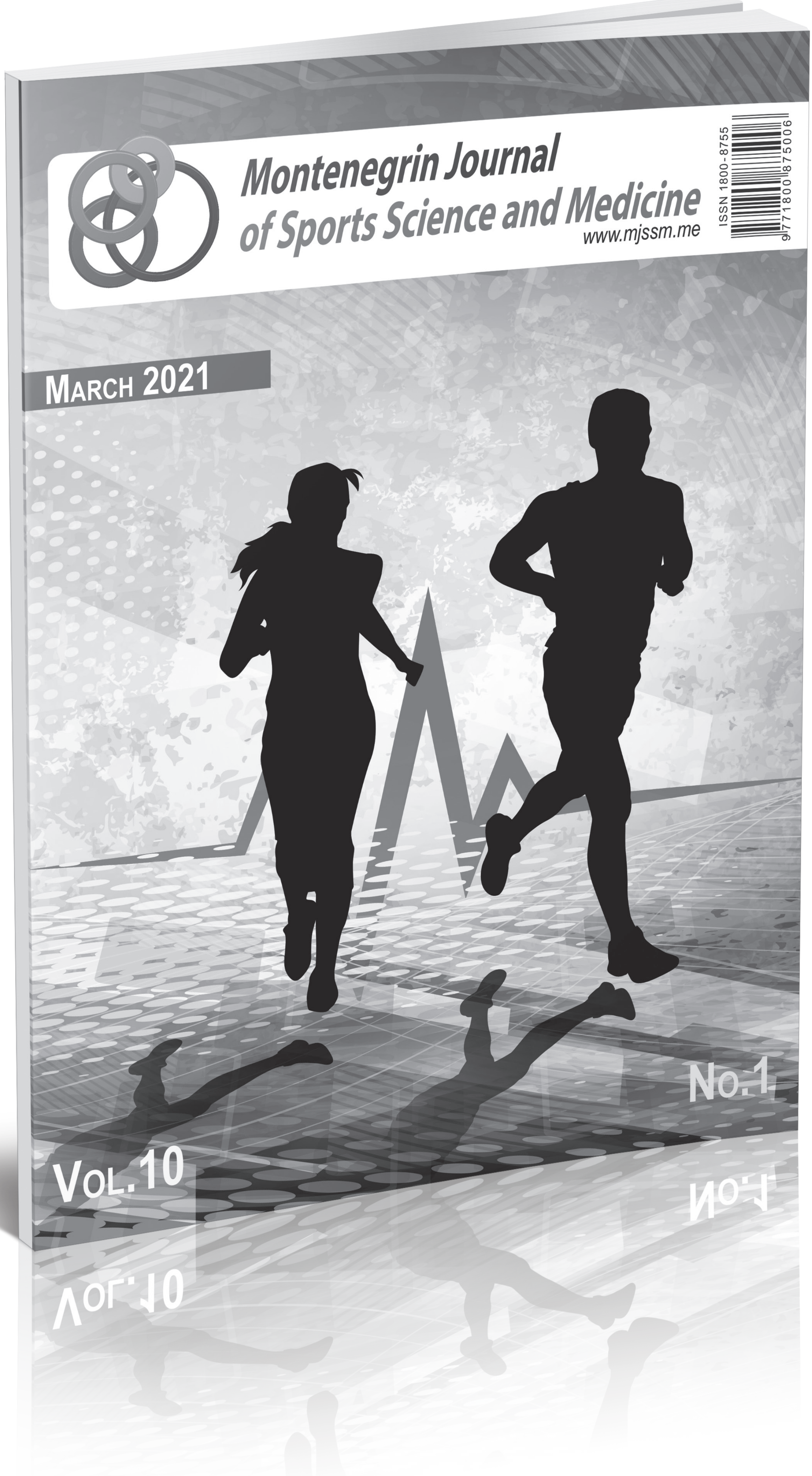

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"Sport, Physical Activity and Health: Contemporary perspectives"

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CRNOGORSKA SPORTSKA AKADEMIJA

MONTENEGRIN SPORTS ACADEMY

7th - 10th April 2022,
Dubrovnik - Croatia